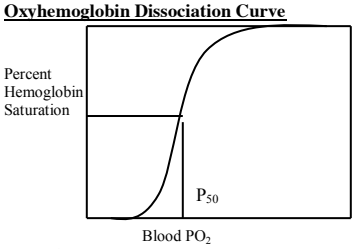
Larynx: (Vagus)
Motor: External Superior Laryngeal = Cricothyroid
(stim of SLN = laryngospasm)
Recurrent Laryngeal = everything else
Sensory: Internal Superior Laryngeal = Above Vocal Cords & Cords
Recurrent Laryngeal = Below Vocal cords
Posterior Cricoarytenoid = Abduct
"You take it out back"
Lateral Cricoarytenoid = Adduct
"You bring it in from the side"
Cricothyroid = Tenses vocal cords
Thyroarytenoid = Relaxes cords

Pharynx:
Motor: Spinal Accessory
Sensory: Glossopharyngeal

Laryngeal Cartilages

Epiglottis	Superior
Thyroid	
Cuneiform (P)	
Corniculate (P)	
Arytenoids (P)	
Cricoid	Inferior

Oxyhemoglobin Dissociation Curve



Normal P_{50} = 26-27 mm Hg

Left Shift	Right Shift
↓ PCO_2	↑ PCO_2
↓ H^+	↑ H^+
↑ pH	↓ pH
↓ Temp	↑ Temp
↑ CO poisoning	↑ 2-3 DPG
(Carboxyhemoglobin)	Sickle Cell
Fetal hemoglobin	
Methemoglobin (Prilocaine, Nitroprusside)	
Smoking	

Right shift unloads O_2 from blood, Right = Released
Left shift increases O_2 affinity - Left = Locked

"Bohr effect" $PaCO_2$ affects Oxyhemoglobin dissociation curve
"Haldane effect" PaO_2 affects CO_2 dissociation curve
"Hamburger shift" Cl^- exchange for HCO_3^- in RBC's:
HCO_3^- out, Cl^- in; non-pulmonary

SaO_2	PaO_2
90%	60mmHg
70%	40mmHg

Dissolved O_2 = 0.003 X PaO_2
O_2 bound to HgB = (1.34 X HgB) (SaO_2)
Total O_2 Content = (0.003 X PaO_2) + (1.34 X HgB) (SaO_2)
O_2 Consumption ≈ 250 ml/min ≈ 3-4 ml/kg/min
Dissolved CO_2 = 0.067 X $PaCO_2$
CO_2 produced & eliminated @ 200ml/min

Control of Ventilation: (brainstem)
Primary Respiratory Centers:
Dorsal Respiratory Group (DRG) Pacemaker
~phrenic & external intercostals } Medulla
Ventral Respiratory Group
~internal intercostal

Secondary Respiratory Centers:
Apneustic Center
~ deep & prolonged respiration } Pons
Pneumotaxic Center
~Shuts off respiration (Hering-Bering 1.5L)

Neuro: Central Chemoreceptors in medulla→ ↑ H^+ .PCO^2 in CSF
Peripheral Chemoreceptors: ↓ O_2 < 60 mmHg
- Carotid: Glossopharyngeal ↑ CO_2
- Aortic: Vagus (Afferent) ↑ H^+
- Stretch Receptors: Vagus

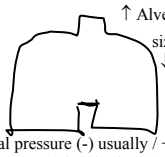
Partial Pressure of CO_2 in CSF = LeChatelier's Principle

Hering-Breuer reflex: Vagus nerve, prevents over-stretching (TV 1.5L)

Inspiration = ↑ **HR** d/t ↓ intrathoracic pressure

	Pulmonary Lung Compliance	Chest Wall Compliance
Peds	↓ (d/t ↓ # alveoli)	↑
Geri	↑	↓

V/Q ↑
PaO_2 ↑
PCO_2 ↓



↑ Alveolar size
↓ V
↓ Q

V/Q = 4/5 = 0.8 [MV/CO]
V/Q = 10/0 = ∞ = dead space
V/Q = 0/10 = 0 = shunt

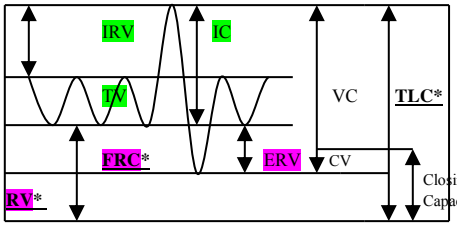
Intrapleural pressure (-) usually / + w/ Valsalva and forced expiration
Intrapulmonary (-) inspiration (+) expiration

Physiologic Dead Space = Anatomic DS + Alveolar DS
***Anatomic DS** = Conducting air passageways + ETT* **≈ 2ml/kg**
Alveolar DS = Ventilation without perfusion
Alveolar ventilation = (TV –DS) x RR
Dead Space = well vent/poor perfusion Shunt= well perfuse/poor vent
↑Dead Space 1. Age, 2. PP Vent 3. PE 4. Lung Dz
Minute ventilation = TV x RR

Compliance = Δ V/ Δ P

	Un-anesthetized		Anesthetized	
Lateral Decubitus	V	Q	V	Q
Non-dependent	↓	↓	↑	↓
Dependent	↑	↑	↓	↑

Jackson-Reese: minimum flow = 5 L/min (2.5-3 X MV)
(Modified Mapleson E)
Bain Circuit min. flow = 70 ml/kg for CV, or 100-300 ml/kg for SV (Modified Mapleson D)
PaO2 = F_iO_2 X 5 PaO_2 = 102 –Age/3 on RA
P_AO_2 = F_iO_2 X 6 PaO_2=100 –(0.4 x age)
P_AO_2-P_aO_2 = (40 X 6) – (40 X 5); if FiO_2 = 40%
CPAP = nondependent (up) *Normal $_{A\text{-}a}O_2$ = 5-15 mm Hg on RA*
PEEP = dependent (down)



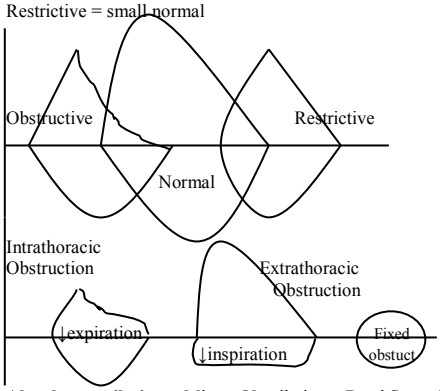
*cannot be directly measured through spirometry

PFT's

Obstructive: **COPD** (hard to get out)	Normal
Asthma, Bronchitis, Emphysema	FEV_1 = 4 L
FEV_1 ↓ < 2.5 L	FVC = 5 L
FVC = normal	FEV_1/FVC = 0.8
FEV_1/FVC ↓ (low) = < 0.7	FEV_{25-75} = 4.7 L/sec

FEV_{25-75} is the best test to assess early stages of COPD-small airway dz
Restrictive: (hard to get in)
Pulmonary fibrosis, Pneumothorax, Scoliosis
FEV_1 ↓
FVC ↓
FEV_1/FVC = normal to high
Moderate risk: If FEV_1 < 2 L & FEV_1/FVC < 50% then more tests, e.g. split-lung function tests. (i.e do not proceed)
High Risk: FEV_1<1, FVC <1.5L or 20m/kg FEV_1/FVC < 35%

Flow Volume Loops:
Obstructive = baby carriage
Restrictive = small normal



Alveolar ventilation = Minute Ventilation – Dead Space Ventil.

Hyperbaric O_2 used to treat:
1. CO_2 poisoning
2. Gas embolism
3. Anaerobic infections (gas gangrene)
4. Decompression sickness (Bends)

CO has a 200-250 X greater affinity for HgB than O_2

4 ways CO_2 is carried in Blood:
1. Physically dissolved (5%)
2. Carbonic Acid (<1%)
3. Bicarbonate Ion (HCO_3^-, 90%)
4. Protein Bound (5%)

CO_2 is 20 X more soluble than O_2
CO_2 produced/min = 200 ml/min or 2.4-3.2 ml/kg/min

Dissolved: O_2 = 0.003 x _____ CO_2= 0.067 x _____

Total CO_2 content of arterial blood = 48 ml CO_2/100 ml blood
Total CO_2 content of venous blood = 52 ml CO_2/100 ml blood

Normal CO_2 arterial-venous difference = 4 ml CO_2/100 ml blood

Carbonic Anhydrase is responsible for converting CO_2 to HCO_3^-

Central chemoreceptors respond to H^+ in the CSF
Peripheral chemoreceptors respond to ↑ H^+, ↑ CO_2, and **↓ PaO_2**

$PaCO_2$ is the **primary** stimulus for ventilatory response

Mast cells can cause bronchoconstriction through histamine release.

6 anatomical characteristics of a difficult intubation:
1.short, muscular neck 2. Receding mandible
3. protruding maxillary incisors 4. Unable 2 visualize uvula
5. limited tempromandibular joint <40mm
6. Limited cervical mobility

5 Contraindications to FOB
1.hypoxia 2. Heave airway secretions
3. bleeding not relieved w/ suction 6. LA allergy
5. inability to cooperate

COPD:
The permanent dilation of a bronchus or group of small bronchi, airway resistance increases, compliance increases
1. Large tidal volume (10-15 ml/kg)
2. Slow inspiratory flow rate
3. Slow respiratory rate
Bronchitis: copious sputum, ↑Hct, "Blue Bloater"
Emphysema: cough w/ exertion, scant sputum, "Pink Puffer"
Post op ventilation: FEV_1/FVC <.50 & Preop CO_2 >50mmHg

Asthma
Wheezing #1
Hypoxia & Hypocarbia w/ alkalosis

Two most common Reasons for Pulmonary Edema:
↑ in Pulmonary hydrostatic pressure
↑ in permeability of alveolar-capillary membrane
Colloid osmotic pressure = 28 mmHg
Hydrostatic pressure = 6-8 mmHg

Causes of ARDS: Right to Left Shunt
1. Shock
2. Fat or air embolism
3. Aspiration
4. Burns
5. Sepsis
6. Trauma
7. Drug ingestion
8. Uremia
9. Pancreatitis
10. Massive blood transfusion
11. Head injury
12. Cardiopulmonary bypass
13. Radiation of thorax
14. Drowning
Hypoxia is the #1 manifestation of ARDS

Five types of Hypoxia:
1. Hypoxic hypoxia (diffusional)
2. Anemic hypoxia (↓Hbg concentration)
3. 2^0 venous to arterial cardiac shunts
4. Histotoxic hypoxia (**cyanide poisoning #1**, toxicity, vitamin poisoning)
5. Hypoxia 2^0 pulmonary disease

S/S of Aspiration: (↑ risk: pH < 2.5, gastric volume > 25 ml)
(Mendelson's syndrome)
1. Wheezing
2. Coughing
3. Cyanosis
4. Pulmonary edema
5. Shock
6. **Hypoxemia- earliest & most reliable sign**
ARDS most serious complication

Four causes of Pulmonary Restrictive Disease:
1. Acute intrinsic restrictive lung disease (ARDS, aspiration, or CHF)
2. Chronic intrinsic restrictive lung disease (sarcoidosis, drug-induce)
3. Chronic extrinsic restrictive lung disease (obesity, ascites, pregna.) "big bellies"
4. Disorders of the pleura or mediastinum

Sarcoidosis
Intrinsic restrictive lung disease
Restrictive ♥ myopathy
↑ Ca^{++}
Splenomegaly
Hepatic granulomas
Optic and facial nerve involvement

S/S of tension pneumothorax:
Hypotension
Hypoxemia
Tachycardia
↑ CVP
↑ PIP
Absence of unilateral breath sounds
Tracheal shift
Asymmetric chest wall movement

Transtracheal Jet Vent:
Cricothyroid membrane

Complications of Transtracheal Jet Ventilation:

Barotrauma & pneumothorax	Sub Q emphysema
Mediastinal air (emphysema)	Exhalation difficulty
Arterial perforation	Esophageal puncture
Damage to tracheal mucosa	***Thickened secretions***

Ventilation modes
spontaneous ventilation : IMV, SIMV, MMV, PSV & HFJV
Do not support spont vent: CMV, AC, and PCV

Respiratory medications
Aminophylline is a phosphodiesterase inhibitor (PDEIII). Phosphodiesterase breaks down cAMP. When phosphodiesterase is inhibited, cAMP accumulates and bronchodilation occurs. It also improves diaphragmatic contractility. Xanthines cause release of NE from sym postganglionic neurons – avoid w/ halothane, adenosine receptors

Caffeine and Theophylline are both methylated xanthines.

Beta-2 receptor stimulation activates adenylate cyclase which converts ATP to cAMP resulting in bronchodilation.

Cromolyn sodium is a mast cell stabilizer that prevents the release of histamine, bradykinin. It prevents bronchospasm in asthmatics, but is not effective once bronchospasm develops. **CHRONIC**

Ipratroprium, a quaternary ammonium compound, is an antimuscarinic used to augment bronchodilation produced by beta-2 agonists. Blockade of the muscarinic receptor leads to a decrease of IP_3 so less calcium is released form intracellular vesicles. Smooth muscle tone is reduced.

Doxapram is a non-xanthine central respiratory stimulant. It increases tidal volume and to a lesser extent respiratory rate. It is not good for newborns because it is dissolved in benzyl alcohol. It acts through peripheral chemoreceptors to stimulate central chemoreceptors.

ETT = Age/4 + 4 (uncuffed) Length = 12 + Age/2
(-) 0.5 cuffed

Wt in Kg	LMA	Cuff vol	ETT	FOB
< 6.5	1	4 cc	3.5	2.7
< 20	2	10	4.5	3.5
20-30	2.5	14	5.0	4.0
> 30	3	20	6.0	5.0
nor adult	4	30	6.0	5.0
lrg adult	5	40	7.0	5.0
lrg adult	6	50	7.0	5.0

*Size is based on weight!

Sterilization temperatures: 275F & 135C
Regurg: head down (#1), disconnect circuit, sx, examine w/ bronchoscope, x-ray, abx (debatable), physiotherapy
Cleaner: Endozime

Differences in Neonatal Respiratory System:
↓ Lung compliance - 2nd to less alveoli
↑ Chest wall compliance-- floppy ribs
↓ FRC ≈ 30 ml/kg-
O_2 consumption 7ml/kg/min (adult- 3.5ml/kg/min)

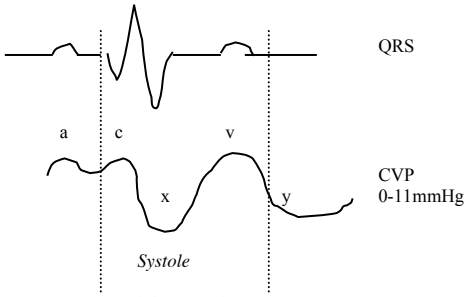
Anterior mandible & thyroid:
Thyro-mental distance > 6.5 cm = ~3 fingerbreadths)

Stress (exercise) ECG has 90% specificity (rules in CAD).

Autoregulation:
Cerebral Perfusion Pressure = MAP 50-150 mmHg
Coronary Perfusion Pressure = MAP 60-160 mmHg
Renal Perfusion Pressure = MAP 80-180 mmHg

Goldman Cardiac Risk Index: (most significant)
Recent MI
S_3 Gallop (CHF)

Four types of abnormal Cardiac wall motion:
1. Hypokinesis (less than normal wall motion)
2. Hyperkinesia (greater than normal wall motion)
3. Akinesis (absence of wall motion)
4. Dyskinesis (paradoxical outward motion)



a = atrial contraction
c = ventricular contraction, tricuspid valve elevation
v = tricuspid closed, systolic atrial filling
x= vent systole, atrial relax, displace tricuspid valve
y=diastole, early vent filling, open tricuspid

RIJ preferred – LIJ has the thoracic duct

Large A waves: tricuspid stenosis, pulmonary stenosis, pulmonary HTN, ↓ right ventricular compliance

Large V waves: tricuspid regurg, r vent papillary muscle ischemia, pericarditis, cardiac tamponade

Multiorifice catheter = 2cm below SVC
Single orifice catheter = 3cm above SVC
T4-T5 interspace above level of 3rd rib

Increased CVP:
Right ventricular failure, tamponade, tricuspid stenosis, tricuspid regurgitation, pericarditis, pulmonary hypertension, chronic left ventricular failure, and hypervolemia

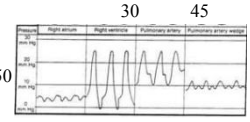
PEEP may read high- need to d/c from vent 10-15 seconds

Venous baroreceptors:
RA & great veins → Bainbridge reflex, stretch of Right Atrium = Increases HR with inspiration via vagus nerve.

Arterial baroreceptors:
Aortic arch → Vagus nerve afferent, stretch
Carotid sinus → Glossopharyngeal (Hering's) nerve afferent
Action potentials (efferent) via vagus and T_1-T_4 sympathetic cardio accelerators.

Swan-ganz Catheters: (1.5ml – max balloon)
Normal pressures & depth of insertion from R IJ

	mm Hg	Depth in cm RIJ +10	RV +15	PA
RA	**1-8**	20	30	45
RV	**15-25**/1-8	30		
PA	15-25/**8-15**	45		
PCWP	6-12	45-50		
LA	2-12			
LV	100-140/0-12			



PAWP is NEVER higher than PADP
In can be in: MS, ↑ alveolar pressure, pulmonary venous obstruc

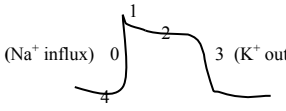
Patient Status	CVP	PCWP
Hypovolemia	Low	Low
Left vent failure	Normal or High	High
Right vent failure	High	Normal
PE	High	Normal
Chronic Pulm Htn	High	Normal
Cardiac Tamponade	High	High

Concentric Hypertrophy = Pressure problem = same size SV
Eccentric Hypertrophy = Volume problem = Larger SV

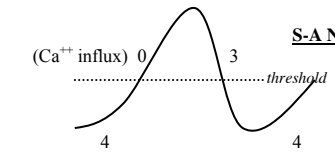
Arterial Line:



1 = anacrotic limb: **contractility & SVR**(ease of LV emptying)
2 = dicrotic limb: blood flow to periphery
3 = dicrotic notch: aortic valve closure, coronary artery perfusion
Pulse pressure greatest in pedis d/t ↑SBP & ↓DBP
High transducer = Low BP Low transducer = High BP
Invasive BP is 20 mmHg higher than non-invasice
Sitting position = transducer @ ear



Vent. Action Potential
0 = Na^+ influx
1 = Cl^- in, K^+ out
2 = Ca^{++} influx
3 = K^+ efflux
4 = Na^+-K^+ pump



S-A Node Act. Potential
0 = Ca^{++} influx and Na^+ (*mostly Ca^{++}*)

SV = CO/HR	60-90 ml/min
CI = CO/BSA	2.5-3.5 L/m/
SI = CI/HR	40-60
SVR = 80(MAP-CVP/CO)	900-1500 dyn
PVR = 80(MPAP-PCWP/CO)	50-150 dynes
$C_oPP = A_oDBP$-PCWP	60-160 mmHg
MAP = DBP + 1/3(SBP-DBP)	80-120 mmHg

RBBB: V_1 = rsR', broad R wave
V_6 = qRS, broad S wave
LBBB: V_1 = no R wave, wide negative S wave
V_6 = no Q wave, wide notched R wave

Inferior, Posterior	II, III, aVF	RCA
Lateral	I, aVL, V_5-V_6	Left Circumflex
Anterior, Septal	V_1-V_4	LAD

Dsyrthmias with MI:
Inferior or posterior: Obstruction of the RCA or circumflex = SA & AV nodal dysrhythmias -sinus arrest, sinus pause, type I AV block, nodal rhythms, & complete heart

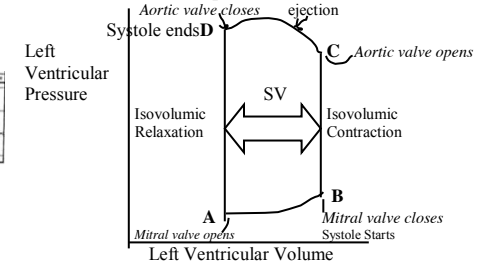
Anterior MI: Wide complex rhythms- bundle branch block, complete heart block, mobitz type II block, idioventricular rhythms.

ECG electrodes: 1 millivolt or 1000 microvolts

Modified V5 lead
place the LL and RA leads in normal positions, then place the **LA** lead over the anterior axillary line at the level of 5th intercostal space & select Lead I as the monitoring lead.

U wave
small positive wave following T wave.
associated w/ ↓ K, ↑ Ca, quinidine, digitalis, epinephrine, intracranial hemorrhage, or papillary muscle dysfunction

LV Pressure Volume Loop:

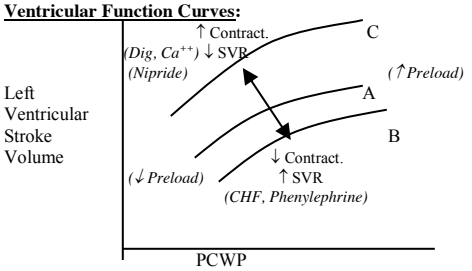


↑ Preload = ↑ SV, same ED vol = (Give fluids)
↓ Preload = ↓ SV, same ED vol = (NTG, Lasix, tamponade)
↑ Afterload = ↑ SVR, ↓ SV, ↑ BP, ↑ ED vol = (phenylephrine)
↓ Afterload = ↓ SVR, ↑ SV, ↓ BP, ↓ ED vol = (SNP)
↑ Contractility = ↑ SV, ↓ ED vol, ↑ BP = (Digoxin, Ca^{++})
↓ Contractility = ↓ SV, ↑ ED vol, ↓ BP = (CHF)

CO:

Liver 27%	Heart 5% (225ml)
Kidney 22%	Lungs 100%
CNS 15% (750ml)	

Ventricular Function Curves:



Aortic Stenosis	Aortic Regurgitation	IHSS (HOCM)
HR↓, SVR ↑, Phenylephrine	Preload ↑, SVR ↓, HR↑ Rheumatic fever	Keep full, SVR ↑ Phenylephrine
Mitral Stenosis	**Mitral Regurgitation**	**Tetralogy of Fallot**
HR ↓, SVR ↑	Preload ↑, SVR ↓, HR↑	Phenylephrine, SVR ↑

PVR ↑ with acidosis & hypercarbia
↑ R→ L shunt

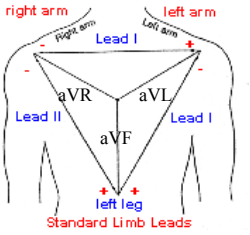
SVR ↓ with acidosis & hypercapnia

Sepsis
↓PCWP ↑ CO ↓ SVR

Becks Triad (Tamponade)
Muffled heart sounds
JVD
HoTN

Statins
Inhibitors of HMG-CoA recluctase
S/E: 1. Liver dysfunction & 2. Severe myopathy

Protamine dose:
1.0-1.3 mg/100 U Heparin



Heparin for bypass:
300 U/kg

Initial dose of FFP:
10-15 ml/kg

Antithrombin III deficiency give FFP→

Autoregulation:
Cerebral Perfusion Pressure = MAP 50-150 mmHg
Coronary Perfusion Pressure = MAP 60-160 mmHg
Renal Perfusion Pressure = MAP 80-180 mmHg

Cranial Nerves & Functions:

#	Name	Function(s)
I	Olfactory	Smell
II	Optic	Vision
III	Occulomotor	Adduction of eye (medial re), pupil size
IV	Trochlear	Eye movements
V	Trigeminal	Chewing, mastication, Facial sensory
VI	Abducens	Abduction of eye (lateral rectus)
V	Facial	Facial muscles, taste (anterior 1/3 tongu
VIII	Acoustic	Balance (vestibular), hearing (cochlear)
IX	Glossopharyngeal	Taste (posterior 1/3), carotid & sinus af
X	Vagus	Parasympathetic efferents, ↓ HR
XI	Spinal Accessory	Motor control of larynx & pharynx
XII	Hypoglossal	Tongue muscles

Oh Oh Oh To Touch And Feel A Girls Vagina- So Heavenly

Cerebral autoregulation MAP 50-150 mmHg
CBF = 750ml/min, 50ml/100g/min, 15% of CO
CBF= CPP/CVR
CPP= MAP-ICP (RAP)- whichever is higher (80-100mmHg)'
Glucose consumption: 5mg/100g/min

Brain Perfusion

	Non Ischemic		Ischemic	
	Blood Flow	Vessel diameter	Blood Flow	Vessel Diameter
Cerebral Steal	↑	↑	↓	0 Δ max dilated
Inverse Steal	↓	↓	↑	0 Δ max dilated

Cerebral steal- Hypoventilation (↑CO_2) & Vasodialators (Nitro)
Inverse- Hyperventilation (↓CO_2)

Orbital Muscles

Superior rectus- supraaduction "look up"	CN III
Inferior rectus- infradduction "look down"	CN III
Medial rectus- adduction "look in"	CN III
Lateral rectus- abduction "look out"	CN VI
Supeiror oblique- look in & down	CN IV
Inferior oblique- look out and up	CN III

Oculocardiac Reflex:
Afferent pathway = Trigeminal nerve- V
Efferent pathway = Vagus nerve - X
ECG manifestation: ↓ HR, Junctional Rhythm, PVC's
Triggered by: traction on extraocular muscles- MEDIAL RECTUS, ocular manipulation, manual pressure on globe
Tx/Blockade: antimuscarinic meds, retrobulbar block, IA

ICP:
Normal = 5-15 mm Hg
Focal ischemia – 25-55 mmHg
Global Ischemia > 55mmHg

Intracranial Transmural Pressure = MAP-ICP

IC Volume:
80% Brain matter & intracellular H_2O
12% Blood
8% CSF

ICP Waveforms:
A waves = plateau waves, found in pts with ↑ ICP
B waves
C waves } lesser magnitude

Twelve S/S of ↑ ICP:
1. Headache
2. N & V
3. Blurred vision
4. Unilateral pupil dilation
5. Papilledema
6. Cranial nerve III (oculomotor) paralysis, ∅ adduct
7. Cranial nerve VI (abducens) paralysis, ∅ abduct
8. **HTN**
9. **Bradycardia** } Cushing's Triad
10. **Irregular respirations**
11. Altered LOC
12. Seizures

Treatment of ↑ ICP:
1. Dehydrate the brain with Mannitol (0.25-1g/kg) or Lasix
2. Give steroids- slowest but may restore BBB
3. **Hyperventilate to $PaCO_2$ 25-30 mm Hg** (1/2 life 6 hours)
4. Restrict fluids
5. Elevate HOB to 30^0
6. Administer cerebral vasoconstrictor (pentathol, etomidate)
7. Control BP
8. Cool pt to 34^0 C for cerebral protection

Posturing:
Decordicate: above cerebellum= flexion, upper & exten- lower
Decerebrate: at brainstem = extension arms & legs, arched body

Cranial Fossae:
Anterior = Frontal lobe
Middle = Temporal lobe
Posterior = Brainstem & cerebellum

Fontanelles	Time to Close
Anterior	18 mos.
Posterior	2 mos.
Anterolateral	2mos
Posterolateral	2 years

Specific gravity of CSF = 1.003-1.009
Hyperbaric- D_{10}
Isobaric- CSF
Hypobaric – NS/Sterile H_2O

CSF forms @ 21 ml/hr or 500-700 ml/day in the choroid plexus

Choroid plexus is located specifically in temporal horn of each lateral ventricle, the posterior portion of the third ventricle, and the roof of the fourth ventricle.
CSF is reabsorbed mostly in the arachnoid villi (4/5), but also in spinal villi & lymphatics.

Total volume: of CSF = 150 ml
CSF Pressure: 10 & 20cm H_2O
Aqueduct of Sylvius is the most common site of CSF obstruction

Four factors governing passage across BBB:
Size - smaller crosses easier
Charge- (ions do not cross- Na, K, Mag)
Lipid solubility- Cross
Degree of protein binding
Water & Gases- Cross
H_2O soluble drugs & Proteins- NO
The Chemoreceptor trigger zone (CRTZ) and the area surrounding the posterior pituitary have no BBB.

Four electrolyte disorders that ↓ Seizure threshold:↑ Sz activity
1. Hypocalcemia
2. Hypomagnesemia
3. Hyponatremia
4. Hypernatremia

Conditions & Medications Likely to ↓ Sz Threshold:
Hypoglycemia
Alkalosis
Demerol is the opioid most likely to cause seizures
Aminophylline & ketamine together

Acute Spinal Shock:
Hypotension occurs due to sympathetic blockade and **bradycardia** due to blockade of cardiac accelerators.
Autonomic Hyperreflexia- T5 or T6

Cerebral Vasospasm:
4-12 days post op
S/S: worsening headache, confusion, HTN

Triple H Therapy:
For treatment of cerebral vasospasm
-Hypervolemia = CVP > 10 mm Hg, PCWP = 12-20
-Hypertension = SBP 160-200 mm Hg
-Hemodilution = Hct 33%
-Medication - Nimodipine

Wake-up test monitors the anterior (ventral) spinal cord, which is supplied by the anterior spinal arteries. These are motor tracts.

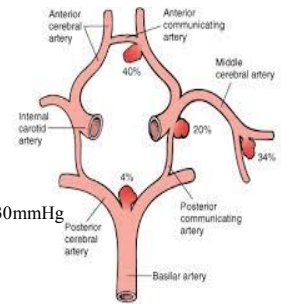
Complications of wake-up test:
1. Recall
2. Extubation
3. Dislodgement of spinal instrumentation
4. VAE from spontaneous ventilation

In Parkinson's avoid
Reglan
Compazine
Droperidal
All meds that ↓ Dopamine

VAE Doppler @ RA 3rd-6th ICS, R of sternum
1. Notify surgeon so they can flood the field or pack
2. Turn off N_2O
3. Administer 100% O_2
4. Aspirate central venous catheter to remove air
5. ↑CVP (Valsalva maneuver)
6. CV drugs to support circulation
7. Bilateral jugular vein compression
8. PEEP
9. Position- left lateral decub w/ 15 degree head down tilt

CSF:
Choroid Plexus
↓
Lateral Ventricle
↓
Foramen Monroe :
↓
3rd Ventricle
↓
Aqueduct of Sylvius
↓
4th Ventricle
↙ ↘
Foramen Luschka Foramen Magendie
↘ ↙
Subarachnoid Space
↓
Brain
↓
Arachnoid Villi

Circle of Willis:
R/L Internal Carotid
Basilar & Vertebral Arteries



Slack Brain
Mannitol/ Diuretics
Hyperventilation $PaCO_2$ = 25-30mmHg
Hypertonic Saline
Head up position

Hypotension
MAP 60-70mmHg MAP 50mmHg
Maintain or ↓ transmural pressure
Results in luxury perfusion –intracerebral steal syndrome

EEG
0.1 millivolts (mV) 100 microvolts (V)
Brain waves ≈ 10-100 microvolts

Delta: 1-4 Hz- greatest amplitude- sleeping adult, abn in wake
Theta: 4-8 Hz- Amplitude- higher than alpha & beta, but lower than delta
Alpha: 9-14 Hz- higher in amplitude, alert but relaxed- eyes close
Beta: 15-40 Hz- low amplitude, frontal head, business activity
Variations seen w/ benzo & propofol – mu wave

Amplitude: Delta >Theta >Alpha >Beta
Frequency: Beta > Alpha > Theta > Delta

Gamma: high-order activity like problem solving (> 25yo)
Mu: beta wave variant- seen over motor areas- amplitude ½ of beta
Lambda: awake patient that is staring, reading or looking @ objects

GA: ↓ high frequency in Beta waves
↑ low frequency in delta & theta waves

Surgical stimulation or light anesthesia: ↑ high frequency, low voltage activity

Cerebral compromise & deep anesthesia: low frequency, high voltage activity

Sevo & Enflurane: can accentuate epileptic activity
Isoelectric 1.5-2.0 MAC
Barbiturates, etomidate, and propofol = burst suppression
Ketamine, opioids and etomidate- **do not** produce a **Δ** in latency & amplitude

Spinal Cord:
Sensory – Afferent – Dorsal Horn S.A.D -Posterior
Motor – Efferent – Ventral Horn Anterior
MMEP: Peripheral- popliteal, Central- anterior
Preganglionic SNS – Intermediolateral Horn

Dorsal-Lemniscal (Sensory): Damage ↓amplitude
SSEP Monitoring (posterior spinal arteries) ↑ latency
- Touch, pressure, vibration
- Dorsal (posterior) cord – Cuneatus & Gracilis tracts
- Ascend ipsilateral side
- Decussate @ brainstem to contralateral thalamus & sensory cortex
- Also goes to RAS where it percolates to sensory cortex
- Somewhat sensitive
- Tibial – electrodes midline scalp, Ulnar- electrodes lateral

Visual evoked potential- CN II- very sensitive to IA
BAEP – CN VIII- barely sensitive (altered most by temp)
Ketamine, etomidate, & opioids, barbs, propofol = no Δ in latency or amplitude in SSEP

Ascending Pain (Anterolateral):
Lateral Spinothalamic Tract (neopalatine)
A-δ Fibers – Myelinated, Fast "first" Pain & temp
Rexed's lamina I & V, dorsal horn
Neurotransmitter - glutamate
C Fibers – Unmyelinated, Slow "dull" Pain & temp
Rexed's lamina II (*substantia gelatinosa*) & *III*, dors
Neurotransmitter – substance P
Interneurons go from II & III to V
Epidural steroids
Both fibers ascend or descend in the tract of Lissauer 1-3 segmnt
Both fibers decussate and ascend on the contralateral side.

Five factors that alter the latency and/or amplitude of SSEP:
1. ↓ Cerebral perfusion 2^0 hypotension, ↓ $PaCO_2$, ↑ ICP
2. Cerebral hypoxia
3. Hypothermia (MOST)
4. Hyperthermia
5. Hemodilution; Hct < 15%

Descending Pain (Dorsolateral):
Dorsolateral Funiculus – modulates pain (spinal analgesia)
Originate in the periventricular and periaqueductal gray areas and terminate on **enkephalin-releasing** interneurons in *Rexed's lamina II* (*substantia gelatinosa*). This inhibits the release of substance P. (*Presynaptic inhibition*)

Periventricular Gray / Periaqueductal Gray } Substantia Gelatinosa } Enkephalin Interneurons } ↓Substance P
All via **nucleus magnus raphe** in the pons and then descending via the **dorsolateral funiculus**
Interneurons = PAG & nucleus raphe

Pain categories:
Nociceptive: physiologic
carried by A-delta-sharp, prickly & C fibers-dull
Somatic-sharp & well localized
Visceral-diffuse, dull & vague
Neuropathic:
Caused by abnormal processing of painful stimuli. Neuropathic pain may occur after injury to neural tissue secondary to systemic disease, infection, trauma, ischemia, deficiencies in metabolism or nutrition, or exposure to environmental toxins or neurotoxin medications.

Lateral spinothalamic (neo) tract- most important spinal tract for pain
Allodynia: perception of an ordinarily non-painful stimulus as pain
Hyperpathia is a combined disorder consisting of hyperesthesia, allodynia, and hyperalgesia

Neurochemical mechanisms of pain involve different NT:
Sub P, Bradykinins & serotonin released → arachononic acid released = thromboxane, prostaglandins & leukotrines

Preganglionic Parasympathetic Nerves originate:
Cranial nerves III, VII, IX, & X 3,7,9,10 craniosacral
Sacral segments S_2-S_4

Weak Acids: (Thiopental, other barbit, [+ Charge/ Na^+, Mg^{++}], Proton Donor
Acid + Acid = unionized

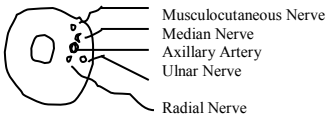
Weak Base: (LA, ketamine, opiods, benzos) [- Charge/ Cl^-, SO_4^-]
Proton Acceptor
Base + Base = unionized

Ionized = H_2O soluble
Non-Ionized – lipid soluble (crosses BBB)
Potency = lipid solubility
Duration = protein binding & solubility
Speed of Onset = pK_a
Fetus pH < maternal pH = ↑ ion trapping

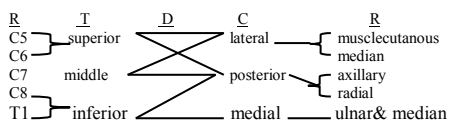
mcg/ml	S/S of Lidocaine Toxicity
3	Circumoral Numbness(non CNS- d/t extracellular extravasation)
4	Lightheadedness
6	Visual Disturbances
8	Muscular Twitching
10	Unconsciousness
12	Convulsions
15	Coma
20	Respiratory Arrest
26	Card iovascular Collapse (widen QT precedes)

Blood flow highest to lowest- loss of LA d/t vascular reabsorb
In Intravenous
Time Tracheal
I Intercostal
Can Caudal
Please Paracervical
Everyone Epidural
But Brachial Plexus
Susie & Spinal
Sally Subcutaneous

Brachial Plexus:



"Robert Taylor Drinks Cold Beer" = Root, Trunk, Division, Cord, Branch



Four Approaches: supraclavicular, infraclavicular, axillary, interscalene

Axillary: for forearm & wrist, safest, *miss the muscultaneous* 30-40ml,
musculocutaneous = 3-5 mL of LA into coracobrachialis muscle.
Supraclavicular = *greatest risk of pneumo*, most compact 40ml
Less likely to miss the peripheral or proximal branches
Interscalene = shoulder surgery, *miss of ulnar nerve* & targets TRUNKS, no hand 40 ml
High incidence of ipsilateral hemidiaphragmatic paresis

Shoulder & humerus = interscalene and Supraclavicular

Cervial Plexus Block:
C1 (motor), C2, C3, and C4 = 4ml
some plastic surgery procedures, carotid endarterectomy tracheostomy and thyroidectomy.
Complications: hiccups, Horner's, hoarse,

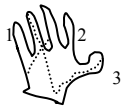
Horners Syndrome= blockage of stellate ganglion @ C7
Least likely w/ axillary block
Ipsilateral side (same)
1.Ptosis (droopy eye lid)
2. Miosis (pupil constriction)
3. Facial & Arm flushing (d/t vasodilatation)
4. ↑ Skin Tem
5. Anhydrosis (lack of sweating on face)
6. Nasal Congestion

Bier Block
Minimum tourniquet time = 15-20 mins or 20-40
Lidocaine 0.5% or Prilocaine 0. 5% -40 – 50ml
No bupivacaine- ♥ tox or chloroprocaine- thrombophlebitis
300 torr or 2.5 times the SBP
Contraindicated: severe crush injuries, uncontrolled hypertension
Raynaud's disease PVD, Homozygous sickle cell

Right Hand: Pronated Supinated

1 = Ulnar
2 = Median
3 = Radial





Nerves that Flex the Forea
Musculocutaneous
Radial

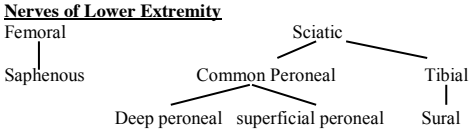
The Radial Nerve innervates
Extension @ elbow, supination of FA, extension of wrist & fingers
Damage = inability to ABDUCT thumb & wrist drop

The median nerve innervates:
Pronation of FA, flexion of wrist
To thumb, index finger, middle finger & lateral ring finger
Innervates the medial aspects of FA
Pronator teres
Flexor carpi radialis
Palmaris longus
Flexor digitorum superficialis
Damage = inability to ADDUCT thumb & Ape Hand

The ulnar nerve innervates:
Flexion of wrist, adduction of all fingers
The little finger & medial ring finger (C8)
In the forearm:
Flexor carpi ulnaris
Medial ½ of flexor digitorum profundus
And in the hand:
Palmaris brevis muscle
Abductor digiti minimi
Flexor digiti minimi
Damage = Claw hand
Innervates the adductor pollicis of the thumb

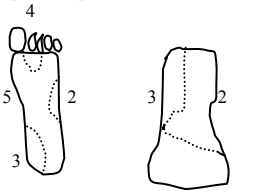
Musculocutaneious
Flexion @ elbow

Nerves of Lower Extremity



Right Foot:

1 = Tibial Sciatic
2 = Sural
3 = Saphenous Femoral
4 = Deep Peroneal Sciatic
5 = Superficial Peroneal



top of foot
Bottom of foot from hee

Functions of nerves of ankle & Foot

1.Saphenous = anteromedial foot, medial anterior calf and the dorsum of the foot
2. Deep peroneal nerve= toe extension & sensation to medial ½
3. Superficial peronal nerve = sensation superficially to dorsum of foot & all 5 toes

4. posterior tibial – sensation to heel, medial sole & lateral sole
5. Sural – sensation to lateral foot

Flexion of foot= medial plantar & lateral plantar - tibial
Extension – peroneal nerve

Superficial : saphenous, superficial peroneal, sura "S's"

Femoral Nerve
L2, L3, L4
Anterior thigh & knee
Anterior muscles of the thigh
NAVEL (nerve, artery, vein, empty space, and lymphatics

Obturator nerve
Provides sensation to the medial aspect of the thigh and motor innervation to the adductor muscles located in the medial thigh

On or above knee surgeries: Femoral, Sciatic, Lateral Femoral, cutaneous obturator

Sciatic
L4, L5, and S1-S3
sciatic nerve innervates the muscles of the back of the thigh (biceps femoris, semitendinosis, semimembranosus, and adductor magnus). As the sciatic nerve continues, it innervates the muscles of the lower leg and foot
Popiteal Block = sciatic

Nerve Injuries
Primary mechanisms responsible: peripheral nerve injury
are transection, compression, stretch, and kinking
Face mask ventilation – CN 5 & 7 (facial & tongue numbness)
LMA – SLN or RLN
Intubation – RLN, SLN, CN 10, CN 12

Ulnar nerve
Is the most commonly injured peripheral nerve
in patients undergoing anesthesia
More common in those with BMI > 38 & men

Brachial Plexus
Placement of shoulder braces = acromion
Extreme flexion at the thigh can result in injury to the sciatic, obturator, and femoral nerves.

Radial Nerve
Damaged = loss of the ability to supinate the extended forearm, wrist drop, abduct thumb, extend the metacarphophalaneal joints

Common Peroneal nerve.
Most commonly injured nerve of lower extremity
Most common injured nerve during *lateral position*
3 issues with common peroneal nerve injury
Loss of dorsiflexion of the foot is consistent with injury to the
Foot drop and inability to evert foot

Pudendal nerves
Fracture table

Sciatic
Protect w/ pillow under knees
Injured when patient rotated to semi supine (hips)

Saphenous
Inside of knee (litho with strap medially)
Numbness & tingling along medial aspect of the calves

Femoral
Decreased sensation LATERAL thigh

Foot drop: Sciatic (lumbosacral, common peroneal), anterior tibial
Hip vag delivery lateral decub feet plantar flex

Complications of retrobulbar block
Stimulation of the oculocardiac reflex, retrobulbar hemorrhage, circumorbital hematoma, penetration of the globe, optic nerve trauma, optic nerve sheath injection, extraocular muscle injury, intra-arterial injection

Transtracheal
Blocking of RLN through cricothyroid membrane w/4% lido
Absorbed across mucous membranes (sim to sublingual)

Regional
High points in spinal canal: High = C3 & L3 Low= T6 & S2
Widest point in space: L2
Biggest vertebral opening: L5-S1
Blood supply to SC – **single** anterior spinal, **paired** posterior
Site of action: nerve root (epidural space), nerve rootlets(spinal), spinal cord
Apnea d/t Hypoperfusion of resp centers in medulla
CSF (+) Epidural (-)
Isobaric = CSF
Hyperbaric = Dextrose solution
Hypobaric = sterile H_2O

Epidural
1-2ml of LA per segment for epidural block
13cm needle
17 or 18 gauge needles
~ 5cm from skin to epidural space (up to 8cm obese)

Batson's plexus in the epidural space communicates with the azygous system- important during times of engorgement which can cause engorgement of the vessels during instances of increased abdominal pressure

Caudal
Sacrococcygeal membrane (injected into epidural space)
Anatomical landmarks: 2 sacral cornua, the coccyx, and the posterior superior iliac spines
Complications: pain at site #1, urinary retention, infection
Dosages: (Adult) Childrens
S5-L2: 15-25ml 0.5-1.0ml/kg of 0.125-0.25
S5-T10: 35ml bupivicaine
premature infant: chloroprocaine 1mlg/kg bolus & 0.3ml/kg
1mcg/kg clonidine

Caudal Dose Bupivacaine:
0.5-1.0 mg/kg
Infant test dose = 0.5 mcg/kg epinephrine
Max dose is 3mg/kg – (bupivacaine)

Needles
Cutting: Quinke, Pitkin
Non-Cutting: Whitacre, Spotte, Greene
SAB 24-27G
Epidural- 18 to 16G Toughy
Nerve Blocks – 23G

Passage of Needle
Skin
Subcutanous tissue
Supraspinous ligament } will not pass through on paramedian
Interspinous ligament } approach (paraspinous muscle)
Ligamentum flavum
Epidural Space
Dura
Subarachnoid

Dermatome Landmarks
C1- Completely Motor
C4- Clavicle T10- umbilicus
T4- Nipples L_4-L_5-Tibia
T6- Xiphoid S_2-S_5-Perineium

The tip of the 12th rib corresponds with L1
The origin of the scapular spine corresponds with T3
The most protuberant cervical vertebra is at the level of C7
The tip of the scapula corresponds with T7
level of the posterior superior iliac spine S2

SAB vs. Epidural
SAB segmental spread is 1^0 d/t: mg, baricity, positions of LA
Epidural spread is d/t volume of LA

Neuroaxial Opioids
1. Hydrophilic: Morphine
Slow onset & prolonged DOA
Intrathecal – 0 early respiratory depression
+ late resp depression d/t rostral spread (6-12 hrs)
Epidural – + early respiratory depression after 2 hours
+ late resp depression d/t rostral spread (6-12 hrs)
2. Lipophilic: Fentanyl, Sufentanil, Alfentanil
Fast onset & short DOA
Intrathecal - + early resp depression (2 hrs)
0 late respiratory depression
Epidural- + early resp depression (2 hrs)
0 late respiratory depression

Four common side effects of intrathecal opiods:
1. Pruritus (most common)
2. Urinary retention
3. N & V
4. Respiratory depression

Common side effects of epidural opioids:
1.urinary retention (bup/morphine)
2. pruritus (morphine)
3.weakness of hands & HoTn

C_8-T_1 = Stellate Ganglion- if blocked = Horner's syndrome
Horner's syndrome- ipsilateral miosis, ptosis, enopthalamos, flushing, ↑ skin temp, anhydrosis, nasal congestion

High Spinal:
C_8 = numbness @ little & ring finger
C_7 = numbness @ middle fingers
C_6 = numbness @ thumb & index finger

Blocks for SAB:
Sympathetic = 2-6 dermatomes higher than sensory
Motor = 2 dermatomes lower than sensory
Progression of blockade: Autonomic>sensory>motor

1. Temperature sensation
2. Proprioception (kinesthetic sense)
3. Motor function
4. Sharp pain
5. Light touch

Type B > Type Aδ = Type C > Aβ > Aα
C type = more resistant to blockade than A & B fibers

Sensitivity: large mylenated > smaller mylenated > unmylenated

SAB additives
Epinephrine 0.2 to 0.3 mg
Clonidine 75 to 100 mcg
phenylephrine 2 to 5 mg
prolong the duration w/o resulting significant ♥ changes

Epidural Steroid Injection
Epidural steroid injections provide relief from acute radicular pain when the nerve root(s) exhibits: edema, inflammation and ↑ levels of phospholipase A_2 expression

Affects unmylenated C fibers

Cervical @ C6-C7 & C7-T1 d/t the largest interlaminar distances

Lumbar sympathetic block @ L2

Methylprenisone 40-120mg & triamcinolone diacetate 40-80mg

Procedures & Level of Block:
TURP – T 10 C-Section T4
ESWL- T4-6 Testicles- T10
Urinary bladder- S2-S4 Tourniquet- T8
Lower abd – T6 Upper abd- T4
Kidney – T10-L1 Cysto- T8-T10
Uterine – T8-T10 hysteroscopy- T10
Hip Arthroplasty- T10

Conditions that Increased height of spinal block
↑ abdominal pressure or engorgement of epidural veins:
pregnancy, ascites, abdominal tumors, kyphoscoliosis,
↑ age = ↓ CSF volume & increase height of spinal blockade

Complications
Infection: streptococci- spinal , staphylococcus- epidural
Failure of block
Backache #1
Spinal Headache #2
quicker onset = greater dural damage- harder to treat
usually the next day
↑ Women > men & Young > old
Tx: bed rest, caffeine, fluids, epidural blood patch (max 20ml)
Neurologic dysfunction allergic reaction
anterior spinal artery syndrome trauma
drug toxicity infection hematoma
total spinal blockade – s/s = dyspnea, resp arrest, HoTN

Arachnoiditis: inflammatory disorder of arachnoid mater which surrounds the spinal cord and cauda equina.
caused by exposure of the arachnoid membrane to povidone iodine solution, vasoconstrictors, LA, blood, and contrast media.

Cauda equina syndrome:
s/s:lower back pain, sciatica, motor & sensory loss, & bladder & bowel dysfunction
d/t trauma, lumbar disc disease, ankylosing spondylitis, tumors, or abscesses in the lumbar area. It has also been associated with prolonged exposure of the cauda equina to high doses or high concentrations of LA that cause direct neurotoxicity.

TNS: transient radicular irritation, pain in the lower back or buttocks that may radiate to one or both legs after a spinal anesthetic
Mepivicaine & lidocaine implicated & lasts ~ 1 week

Absolute Contraindications to Regional Anesthesia
1.Infection @ site 2. Coagulopathy
3. Marked hypovolemia 4. True allergy to LA
5. Pt. refusal/inability to cooperate 6. Severe Stenosis
7. ↑ ICP 8. Abruption placentae

Relative Contraindications to Regional Anesthesia

1.Preexisting neurological dz 2. Back disorder (Ankylosis)
3. Heart Disease 4. Surgery above umbilicus
5. Failure to obtain free flow 6. Sepsis
7. Mobitz type I or II 8. 3rd degree w/o paceer

Risk & Complications of Regional Anesthesia

1.infections 2. HoTN 3.Urinary retention
4.inadequate analgesia 5. Intravascular injection
6. High spinal 7.PDPH 8. Back pain
9.N/V 10. pain on injection

Parturient & LA

SAB & Epidural = ↑ spread & Depth
Related to ↓ thoracolumbar CSF volume & ↑ neural sensitivity
Hormonal progesterone in CSF may ↑ segmental spread
Bupivacaine 0.25-0.5% = good sensory but minimal motor

Anticoagulants

NSAID, ASA, sub-q or mini dose heparin- No issues
IV heparin – need nml PTT before regional
Hold 1 hr after placement
Cathetars removed 2-4/hr AFTER last heparin dose
Heparinization 1 hr after catheter removal
LMWH- first dose 24 hrs post op (2x daily dosing)
6-8 hs post op (daily dose)
First dose 2 hours after catheter removal
Warfin- stop 4 days before surgery & INR < 1.5
Fibrinolytic or thrombolytic – 10 days
Ticlodipine – 14 days
Clopidorgrel- 7 days
GPIIb/IIIA – hold for 4 weeks post operative

↑ potency of LA:

↑ potency = ↑ protein binding, ↑ DOA, ↑ affinity for Na channels, ↑ tendency of cardiac toxicity

↑ # of carbons, + Halide, + ester linkage, large alkyl on tertiary amide

Bier Block:

Method of anesthetizing a limb by IV injection while blood flow to extremity is occluded by a tourniquet
Minimum: 15-20 mins (don't release before- local in systemic)
Max: 40-65 min (usually d/t tourniquet pain)

Local anesthetics:

Cmin: minimum concentration of LA to produce conduction block
MOA: non-ionized LA crosses membrane and ionized binds to receptor to produce effect. (Ca controls Na permeability)
Block rapidly firing nerves > idle nerves
Autonomic > perception of pain/touch/temp > motor > proprioception
+ Na^+ Bicarb =↑ speed of onset, intensity of block,↓ pain, ↑ DOA
+ Dextran = ↑ DOA
+ hyaluronidase = ↑ spread of the LA into the tissue.

Ester local anesthetics are eliminated by plasma pseudocholinesterase except cocaine, which is eliminated by hepatic metabolism.
Metabolism: chloroprocaine > procaine > Tetracaine
↑ likely hood of allergic reactions d/t para-aminobenzoic acid

Tetracaine is hydrolyzed much more slowly by plasma cholinesterase and is highly protein bound. Therefore it is the most toxic ester local anesthetic. **Procaine** is the least potent ester.

Cocaine is an ester of benzoic acid and is the only local anesthetic that produces vasoconstriction. It is also the only naturally-occurring local. Blocks epi uptake: caution in use of tricyclics, MAO's, catecholamine metabolism blockers. The max dose is 1.5 mg/kg
Cocaine & ropivicaine = constriction

Amide local anesthetics are metabolized by hepatic metabolism.
Metabolism: prilocaine >etidocaine > lido > Mepivicaine >Bup
Prilocaine is the least toxic amide LA.

Prilocaine is metabolized to orthotoluidine. **Orthotoluidine** is an oxidizing agent capable of converting hemoglobin to methemoglobin.

Bupivacaine is highly lipid soluble and dissociation form sodium channels are slow. Cardiac toxicity is high.

Mepivicaine, etidocaine, & bupivacaine = no enhancement w/epi

Tetracaine, Etidocaine, and Bupivacaine are about equipotent & most toxic LAs. Etidocaine causes seizures at a lower plasma concentration. Etidocaine & Bupivacaine have lowest maximal dose ranges.

Monoethylglycinexylidide: an active metabolite of Lidocaine that contributes to toxicity even when lidocaine plasma levels are low

Motor Blockade with LA

Minimal: lidocaine 1%, Mepivicaine 1%, Bupivacaine 0.25%
Dense: chloro 3% (most), lidocaine %, mepiviaince 2-3%, etidocaine 1.5%, prilocaine 3%

Four *anticholinesterases* prolong esters: ↓ plasma pseudocholinesterase
1. Echothiophate (irreversible)
2. Neostigmine
3. Pyridostigmine
4. Edrophonium

Four conditions that ↓ plasma cholinesterase:
1. Pregnancy
2. Liver disease (cirrhosis)
3. First six months of life
4. Atypical plasma cholinesterase

Volatile anesthetics, **propranolol**, and cimetidine decrease hepatic clearance of amides. (They inhibit Cytochrome P-450)

Avoid Beta-blockers with amide LAs:
Labetalol & Propranolol
Also:
Digitalis & Ca^{++} channel blockers

Bretyllium is used to treat cardiac toxicity by amides.
Three reasons to add epinephrine to LA:
1. Prolong the duration of anesthesia
2. ↓ Systemic toxicity by ↓ rate of absorption
3. **Permit use of larger amounts of LA**

1% = 10 mg/ml
Epinephrine = 1:200,000 = 5 mcg/ml
Concentration = Amt/Vol C=A/V
Amt = Conct X Vol
% = gm/100 ml

Max Dose Epinephrine:
Subcutaneous or Submucosal infiltration:
2-3 mcg/kg for adults
or 1 mcg/kg on Halothane

3 mcg/kg for children
or 1.5 mcg/kg on Halothane
With local anesthetic:
200-250 mcg
or 3-5 mcg/kg } Adults

*Do not inject epi:in/around end arteries- fingers, toes, ears, penis, nose

EMLA Cream:
2.5% Lidocaine, 2.5% Prilocaine
Biggest barrier = stratum corneum
Psoriasis or broken skin = ↑ onset & ↓ duration (↑ toxicity)
Contact time at least one hour under an occlusive dressing
Reaches a depth of analgesia of about 3-5 mm,
DOA = 1-2 hours.

ESTER	Max Dose (mgkg)	Duration
Benzocaine	Na	
Chloroprocaine	12 (800)	0.5-1
Cocaine	3 (200)	
Procaine	12 (800)	
Tetracaine	3 (200)	1.5-6
AMIDE		
Bupivicaine	3 (175)	1.5-8
Lidocaine	4.5, 7 w/epi (500,700)	0.75-2
Mepivicane	4.5, 7 w/epi (300,500)	1-2
Prilocaine	8 (400)	0.5-1
Ropivicaine	3 (300)	1.5-8

Manifestations of hypersensitivity reactions include:
1. Localized edema
2. Urticaria
3. Bronchospasm
4. Anaphylaxis

Cardiac Toxicity:
Hypoxia, hypercarbia, and acidosis
Tx: Fluid bolus, rest, pain meds, caffeine, blood patch (10-30cc)

TNS: Transient Neurological Symptoms
Lidocaine spinals
Risk Factors: lithotomy, outpatient, knee arthroscopy
Tx: NSAIDs—d/t sensory nature

Lipid Rescue:
20% Intralipid
1.2 to 2 ml/kg and then an infusion
0.25/ml/kg/min for 30-60 mins

Benzocaine:
Ester, Weak ACID
May cause methemoglobinia

Tests:

Sensitivity:
SnNOut
Sensitivity, Negative, Out

Specificity:
SpPIn
Specificity, Positive, In

Volatile Anesthetics:

		VP	Bld:Gas	Oil:Gas	F_A/F_I	MAC	.70 N_2O
	N_2O		0.47	1.4	.99	104	
Sally	Sevo	170	0.65	53.4	.85	2.1	0.66
Eats	Ethr	175	1.9	98.5	.65	1.68	0.60
Ice cre	Iso	239	1.4	90.8	.73	1.15	0.50
Hot	Halo	243	2.3	224	.58	0.74	0.29
Days	Des	669	0.42	18.7	.91	6.3	2.83

Note! There may be additive effects between the cardiac depression associated with Enflurane (or for that matter Halothane) and beta-adrenergic antagonists (i.e. Esmolol)

Small V_d = fast elimination

↑ potency = ↑ lipid solubility = ↓ MAC

Oil/gas: measurement of solubility

Blood Solubility = speed of uptake

Inhalation agents: ↑ solubility = ↓ speed of inhalation induction
↓ solubility = ↑ speed of inhalation induction

Volatile = ↑CBF, ↓CMR
Ketamine/N_2O = ↑CBF,↑CMR
IV anesth = ↓CBF, ↓CMR

Vapor pressure of liquid dependent on SOLEY on temperature

Wrong Agent in Vaporizer :

High	Low	High
Low	High	Low

Percentage of volatiles metabolized:

Halothane	15-20%
Enflurane	2.4%
Isoflurane	0.2%
Desflurane	0.02%
Sevoflurane	3.0%

Cardiovascular Side Effects:
Sux = ↓ HR, Histamine
Mivacurium = Histamine
Atracurium = Histamine
D-Tubocurarine = Histamine, ↑ HR, ↓ BP, ganglionic blockade
Metocurine = Histamine, ↑ HR, ↓ BP, ganglionic blockade
Pancuronium = ↑ HR, ↑ BP
Gallamine = ↑ HR, ↑ BP

Agent	Renal	Biliary	Metabolism
Succinylcholine	Neglig	Neglig	Primary
Atracurium	Neg	Neg	Primary
Mivacurium	Neg	Neg	Primary
Cisatracurium	Neg	Neg	Primary
Vecuronium	Second-20%	Primary	Second
Rocuronium	Second	Primary	Second

Brain uptake of anesthetics depends on:
1. Blood solubility
2. Cardiac output
3. Alveolar ventilation
4. Inspired concentration

Three ways to ↑ speed of equilibrium:
1. ↑ Inspired anesthetic concentration
2. Second gas effect
3. ↑ Alveolar ventilation

Two most important factors for ↑ alveolar partial pressure:
1. Inspired concentration
2. Blood solubility

Partial pressures
Inspired>Alveolar>Arterial blood>Brain
Note! *This order is reversed during emergence when gas is turned off*.

The **Meyer-Overton Theory** explains that the anesthetic potency of anesthetic agents directly correlates with their lipid solubilities.

MAC is the "Minimum Alveolar Concentration" of anesthetic at one atmosphere that produces immobility in 50% of patients exposed to a noxious stimulus. MAC is inversely proportional to potency.

MAC ≈ ED_{50} of non-inhalational drugs. 1.3 MAC ≈ ED_{95}

There is approximately 1% ↓ in MAC for every 1% of N_2O delivery.

Highest Mac 6mos-12mos

Seven factors that ↓ MAC:
1. Increasing age
2. Hypothermia
3. CNS depressants
4. Acute ethanol intoxication
5. Alpha-2 agonists (Clonidine)
6. Pregnancy
7. ↓ Levels of CNS neurotransmitters

Three factors that ↑ MAC:
1. Hyperthermia
2. Hypernatremia
3. ↑ Levels of CNS neurotransmitters

Volatile anesthetics are metabolized in the liver by cytochrome P-450 in hepatic microsomes.

An **oxidative trifluoroacetyl** metabolite of Halothane is thought to be responsible for acute hepatotoxicity in susceptible individuals. Reductive liver metabolism occurs with Halothane in the presence of hypoxia. **Thymol** is the preservative in Halothane.

Fluoride is the most clinically important metabolite of Enflurane.

Inorganic fluoride and chloride are common metabolites of Halothane and Enflurane.

N_2O is the only inhalational agent without a halogen.

Acceptable levels in the OR:
N_2O & Volatile together:
N_2O = 25 ppm
Volatile = 0.5 ppm

Volatile alone:
Volatile = 2 ppm

N_2O is metabolized to N_2 in the intestine by reductive anaerobic metabolism.

Six contraindications to the use of N_2O:
1. Venous air embolism
2. Malignant hyperthermia
3. Ear surgery (middle ear)
4. Closed pneumothorax
5. Potential pneumocephalus
6. Bowel obstruction

Four adverse side-effect of N_2O:
1. Aplastic anemia
2. Congenital anomalies
3. Spontaneous abortion
4. CNS toxicity

↓ methionine synthetase- B_{12} deficiency = no N_2O
N_2O decreases BP and CO when added to high dose opioids.
N_2O ↑ PVR and PA blood pressure due to mild sympathomimetic effects. It will support fire, but is neither flammable nor explosive.

↑ CBF & ↑$CMRO_2$
Three renal changes associated with volatile anesthetics:
1. ↓ RBF
2. ↓ GFR
3. ↓ UO

Halothane least potentiates NDMRs.

Isoflurane and Desflurane most ↓ SVR, Halothane has little effect on SVR.

Halothane and Enflurane produce the greatest myocardial depression.

Halothane and Sevoflurane most depress the baroreceptor reflex. (There is no increase in HR despite decreases in BP)

Isoflurane depresses the temperature-regulating center in the hypothalamus.

Isoflurane, Desflurane, and Sevoflurane decrease cerebral metabolic rate.

N_2O alone increases cerebral blood flow & ICP.

Enflurane and Desflurane most depress ventilation. Halothane least depresses ventilation.

Sevoflurane is most degraded by soda lime and Desflurane least.

Isoflurane facilitates CSF absorption = favorable effect on CSF

Decrease response to CO_2 ventilatory drive

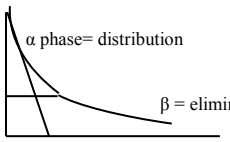
Fluoride: des =6, iso= 5 sevo=7

Steady-state
Point at which the plasma concentration of a drug is in equilibrium with all other tissues is the body

Receptors & Drugs
Agonist: affinity and efficacy
Antagonist: affinity for a receptor but lacks efficacy (cannot produce conformation Δ)
Competitive: can be overcome by ↑ concentrations of agonist
Non-Competitive: antagonism can't be overcome by ↑ concern
Partial Agonist: bind with the receptor and has some efficacy, but it cannot elicit the maximal tissue response
Inverse Agonist: but results in the opposite reaction of an agonist

Zero Order Kinetics
Constant **AMOUNT** of drug over a constant time
ASA, phenytoin, ASA

First Order Kinetics
Constant **FRACTION** eliminated per time
1 compartment = albumin
2 compartments = most other drugs
$V_d = Q/Cp_{t=0}$ Q = quantity of drug injected
$Cp_{t=0}$ = plasma concentration @ time=0
Draw line back from elimination phase to t = 0



Dose response curve:
Potency: determined by the binding affinity of receptors for the drugs as well as the efficiency of coupling of binding to response
Slope: relationship between dose and effect
Efficacy: maximum drug effect

Phase I biotransformation:
Alter the molecular structure of a drug by modifying an existing functional group of a drug.
1. Oxidation
2. Reduction
3. Hydrolysis
-Cytochrome P450 participates in most **oxidation** and some reduction.

Phase II biotransformation:
Consists of a coupling or conjugation of a variety of endogenous compounds to polar chemical groups of the drug.

Biotransformation often makes drugs more water soluble and inactive for excretion in the urine or bile.

Six groups of drugs metabolized by Cytochrome P450:
1. Barbiturates
2. Opioids
3. Benzodiazepines
4. Amide LA's
5. Tricyclic antidepressants
6. Antihistamines

Quantal Dose Response:
Therapeutic Index = LD_{50}/ED_{50}
ED_{50} is the dose of drug that is effective in 50% of patients.
TD_{50} dose that produce toxic effect in 50% of animals
LD_{50} death to 50%

Elimination half-time (T ½) = time taken for the plasma concentration to fall by one-half. T ½ is directly related to V_d and inversely related to Clearance (Cl).

$Cl = V_d / T ½$
↑ Vd= ↑ T1/2 Small Vd=↓ T1/2
Fast CL=short T1/2 Slow CL= Long T1/2

To figure out ½ Lies and amounts excreted

Time	½ Life	Amount of Drug
0	0	0
1	50 (1/2)	50%
2	25 (1/4)	75%
3	12.5 (1/8)	87.5%
4	6.25	93.8%
5	3.125 (1/32)	96.9%
6	1.562 (1/64)	98.4%
7	0.782 (1/128)	99.2%

Second messengers:
Molecules that relay signals from receptors on the cell surface to target molecules inside the cell
cAMP, cGMP, IP_3, calcium

Proteins
Albumin = acid
Alpha-1 acid glycoprotein & Beta-globulins = Base

GABA is the major inhibitory transmitter of the CNS. It opens Cl^- ion channels. It hyperpolarizes neurons inhibiting action potential production.

Barbiturates, benzodiazepines, propofol, and etomidate work primarily on the **GABA receptor**.
Opens Cl^- channel- hyperpolarization

Current research also indicates that inhaled anesthetics also work on **GABA receptors**.

Barbiturates
Prolong the attachment of GABA to its receptor.
They work in the reticular activating system (RAS).
Sodium Thiopental (acid) is 72-86% bound to albumin. It reduces the sensitivity of the central respiratory center to CO_2. It's onset is within 10-15 seconds. It's elimination half-time is 11.6 hours. Metabolized by redistribution dependent on CO.

↓ $CMRO_2$ & ↓CBF (2nd ↑ cerebral resistance)
Inverse Steal
Reconstitute w/ Sterile Saline (NO LR-precipitate)
Hyperalgesia

S/S of intra-arterial Thiopental injection:
1. Arterial vasospasm with intense pain down the arm
2. Blanching of the skin with loss of distal pulses
3. Eventual cyanosis and possibly gangrene

Intra-arterial injection is treated with Phenoxybenzamine (Dibenzyline).

pH of Barbiturates is > 9.0, pH of 10-11 is often cited.

Barbiturates are contraindicated in status asthmaticus and porphyria.

Methohexital is associated with a higher incidence of hiccups than other non-opioid induction drugs.

Benzos: (base)
Sedative: effects: the cortex
amnesia: forebrain and hippocampus
anxiolytic effects:
amygdala, hippocampus, & limbic system.
↓swallowing reflex & upper airway reflexes
↓$CMRO_2$ & ↓CBF
Flumazenil- competitive antagonist of benzos

Propofol (acid)
weak acid
2,6 diisoprorylphenol
↓ SVR = ↓BP
Liver metabolism 70% & lung metabolism 30%
*Caution with soybean & egg allergy

Etomidate: (base)
Maintains CV stability the best.
It directly depresses the adrenal cortex.
It ↓ cerebral blood flow, ICP, & $CMRO_2$

Venous thrombosis and phlebitis are most likely after etomidate, diazepam, and lorazepam.

Four potential problems during recovery from etomidate:
1. Suppression of adrenocortical response to stress
2. N & V
3. ↓ Plasma cortisol concentration
4. Depressed immune response

Ketamine: (base)
Causes dissociation between the *thalamocortical* and *limbic* systems by antagonistic actions on the **NMDA receptors**.
Dysphoria is cause by misperception and/or misinterpretation of auditory and visual stimuli by stimulating the **kappa receptor**, antagonizing the **muscarinic receptor**, and stimulating the **sigma receptor**.

♥ Effects: ↑ MAP, CI, PAP, CVP, HR
Ketamine produces bronchodilitation
↑ airway secretions- give glyco
Analgesia
↓ emergence delirium in kids & higher bioavailability in kids

Opioids: (bases)
Shortest elimination ½ life
Remi < Alfent < MSO_4</= Sufentanil < Meperidine < fent

Potency
Sufent > remifent > fent > alfent > MSO_4 > Meperidine

Morphine:
26-36% is protein bound in the adult
Less is protein bound in the neonate 2^0 ↓ alpha-1 acid glycoprotein.
Metabolite: morphine-6-glucuronide- prolonged in RF & crosses BBB by mass action

Meperidine (Demerol)
↓ myocardial contractility and ↑ HR
↓ shivering - Kappa receptors
↓ sz threshold (↑ having a sz) d/t Normeperidine
It should be avoided with MAO inhibitors & Imipramine

Adverse S/S of MAO inhibitors & Demerol:

1. Hyperpyrexia	**♥ Effects**:
2. HTN	↓ HR,
3. Hypotension	↓SVR
4. Respiratory depression	↓ venous retur
5. Skeletal muscle rigidity	↓ BP
6. Seizures	
7. Coma	

Most clinically used opioids are relatively selective for Mu receptors.

Lamina II- substania gelatinous of SC
PAG & periventricular area in brain stem = spinal analgesia

Spinal analgesia is mediated primarily by **Mu-2** receptors, but also by kappa and delta receptors. Besides the substania gelatinosa (Rexed's lamina II), the periaqueductal and periventricular gray areas are important sites of spinal analgesia.
Supraspinal analgesia is primarily mediated by **Mu-1** receptors, but also by kappa and delta receptors.

Mu-1 receptors produce:
(low abuse potential) (Supraspinal analgesia)
1. Euphoria
2. Miosis
3. **Bradycardia**
4. Hypothermia
5. Urinary retention
6. Pruritus

Mu-2 receptors produce:
(high abuse potential) (Spinal analgesia)
1. Respiratory depression
2. Marked constipation
3. **Physical dependence**

Kappa receptors produce:
(Spinal [Kappa-1] and Supraspinal [Kappa-3] analgesia)
1. Sedation
2. Dysphoria

Four Ventilatory effects of opioids:
1. ↓ Breathing rate
2. ↓ Minute ventilation
3. ↓ Response to CO_2 2^0 brainstem depression
4. ↑ Arterial CO_2 tension

Naloxone, Nitroglycerine, and Glucagon can reverse opioid-induced **sphincter of Oddi spasm**.

Alfentanil is eliminated faster than all other opioids (except Remifentanil) because it has a small Vd. The elimination ½ time is 10-30 minutes.

Remifentanil is metabolized by blood and tissue nonspecific esterases. Has glycine buffer- DO NOT use in neuroaxial.

Agonist-antagonist opioids
Kappa: provide analgesia
Mu: reverse respiratory depression
Naltrexone, naloxone, nalbuphine

Muscle Paralyzation
Eye muscles → extremities→trunk→abd muscles→ diaphragm.
Recovery is restored in reverse order

Facial muscle = diaphragm
Abductor pollis = readiness for intubation
Recovery from NMB = ulnar nerve

MOA NMB:
Site of action is the motor end plate- nicotinic receptors
ALL MR resemble acetylcholine

Four NDMRs that release histamine:
1. d-Tubocurarine
2. Metocurarine
3. Atracurium
4. Mivacurium

Mivacurium is metabolized by plasma cholinesterase. 25% spontaneous recovery is reached in 13 minutes in adults and 7 minutes in children.
Atracurium is eliminated by ester hydrolysis and Hoffman elimination; **Cisatracurium** is only eliminated by Hoffman elimination.

Hoffman elimination is temperature and pH dependent. The rate of metabolism is slowed by acidosis or ↓ temperature.

Laudanosine is a lipid-soluble metabolite of atracurium that can cause CNS stimulation in high concentrations.

Four MRs that use renal excretion least:
1. Succinylcholine
2. Atracurium
3. Cisatracurium
4. Mivacurium

Three NDMRs not significantly excreted by kidneys:
1. Atracurium
2. Cisatracurium
3. Mivacurium

Pipecurium is primarily eliminated by renal (70%) and secondarily by biliary (20%).
Rocuronium undergoes no metabolism, eliminated w/ no Δ

What can augment NMB:
1. Hypermagnesium
2. Hypocalcemia
3. Hypokalemia
4. VA : des > sevoflurane > iso > N_2O/fentanyl
5. Hypothermia

Eleven possible complications of Succinylcholine administration:
1. Hyperkalemia
2. Bradycardia (Succinylmonocholine-metabolite @ ♥ SA Node)
3. ↑ HR and/or BP
4. Skeletal muscle myalgia
5. Allergic reaction
6. Triggering of MH
7. Sustained masseter muscle contraction
8. Myoglobinuria
9. ↑ IOP (NOT prevented with defasculating dose)
10. ↑ Intragastric Pressure (prevented with defasiculating dose)
11. ↑ ICP (prevented with defasiculating dose)

Five conditions that accentuate succinylcholine-induced hyperkalemia:
1. Unhealed third-degree burns
2. Denervation of skeletal muscle
3. Severe skeletal muscle trauma
4. Upper motor neuron injury (head injury, Parkinson's, CVA)
5. Muscular dystrophy
6. Renal Failure w/ hyperkalemia
7. Severe Sepsis
8. Duchennes
9. Guillian Barre

Eleven clinical manifestations of Malignant Hyperthermia:
1. Hypercarbia
2. Tachycardia
3. Tachypnea
4. Hyperthermia
5. Hypertension
6. Cardiac dysrhythmias
7. Acidosis (metabolic)
8. Hyperkalemia
9. Skeletal muscle rigidity
10. Myoglobinuria
11. Hypoxemia

The earliest sign of MH is ↑ $ETCO_2$
Temperature may increase 1-2 C^0 every 5 minutes.
Succinylcholine and volatile agents are triggering agents
Masseter muscle rigidity is an early sign of MH.

-1% of children experience masseter muscle rigidity after halothane and succinylcholine administration. CPK > 20,000 confirms the diagnosis after masseter muscle rigidity following halothane and succinylcholine administration.
- Halothane-caffeine contracture test is the standard diagnostic test for MH, but it has too many false positives.

Eight actions for initial management of MH:
1. Discontinue inhaled agents & Sux
2. Hyperventilate with 100% O_2
3. Administer Dantrolene
4. Treat acidosis with $NaHCO_3$ (1-2 mmoles/kg)
5. ↓ Body temp to 38 C^0
6. Replace anesthesia circuit and CO_2 absorber
7. Monitor $ETCO_2$ & ABGs
8. Treat hyperkalemia and dysrhythmias if necessary

Dantrolene: binds to the ryanodine-1 channel and inhibits the calcium channel in the sarcoplasmic reticulum. Decreasing the release of Ca^{++} from the sarcoplasmic reticulum in skeletal muscle and causes skeletal muscle to relax. The initial dose is 2.5 mg/kg followed by 1-2 mg/kg boluses to a maximum dose of 10 mg/kg. The therapeutic blood level is 2.5 mcg/ml. Vials of Dantrolene contain 20 mg and each is mixed with 60 ml of sterile distilled H_2O.
It should be repeated every 10-15 hours for three days.

Five complications include:
1. Reoccurrence
2. DIC
3. Myoglobinuric renal failure
4. Skeletal muscle weakness
5. Electrolyte abnormalities

Gastric lavage is the best method to ↓ Temp with MH.
Procainamide, 15 mg/kg is the best antiarrhythmic for MH.

The incidence of MH in children is 1:15,000
The incidence in adults is 1:50,000
The mortality rate is 10%

Neuroleptic malignant syndrome can mimic MH, but the onset and recovery are different. Patients treated with antipsychotic drugs such as Haldol, prolixin, or thorazine are susceptible to neuroleptic malignant syndrome. **Fever** is the cardinal sign

Anticholinergic:
1. Atropine, - most ↑ HR
2. Scopolamine- most sedative
3. Glycopyrrolate - does not cross BBB d/t being a quaternary

Combine reversibly w/ muscarinic cholinergic receptors prevent acetylcholine from binding to the receptor.
Sedative effect: Scopolamine > atropine > 0 glycopyrrolate
Antisialogogues effect: Scopolamine > glycol > atropine
HR: Atropine > glycopyrrolate >scopolamine - < PR interval
Infants & elderly little effect on HR
Bronchodilatory effects: Ipratropium
Do not use scopolamine in GLAUCOMA

Gastric Effects: ↓gastric secretions, ↓ peristalsis and intestinal motility, ↑ gastric emptying time, & ↓ lower esophageal sphincter tone.
Central anticholinergic syndrome:
Scopolamine & atropine both cross the blood-brain barrier and block muscarinic cholinergic receptors in the CNS, producing restlessness, hallucinations, somnolence, and potentially, unconsciousness. Predisposed patients: Tricyclic antidepressants (like amitriptyline), antipsychotics, and antihistamines (antimuscarinic characteristics)
Tx: physostigmine

Xanthines:
Aminophylline & theophylline
cause release of norepinephrine from sympathetic postganglionic neurons and should be avoided with Halothane.

Halothane should be avoided with patients intoxicated with cocaine or using imipramine, because they both block reuptake of norepinephrine.

Calcium channel blockers and volatile agents act synergistically

Chemotherapy Medications and Site of Toxicity

Bleomyocin – Lungs	doxorubicin -Heart
Cisplatin- Kidneys	cyclophophains, streptozocin, Methotrexate-Liver

Immunosuppressive

3 major risks- 1. Infection 2.malignency 3. Progressive vascular dz
Calciumium inhibitors = cyclosporines & tacrolines
S/E: HTN, hyperlipidemia, ischemic valve dz, DM, neuropathy, CNS effects

Calcium Channel Blockers work:

Phase 2, plateau phase of ventricular action potential
Phase 4 of the pacemaker action potential

Drugs to avoid with MAO inhibitors:

1. Tricyclic antidepressants (imipramine)
2. Opioids (especially Demerol)
3. Indirect acting sympathomimetics (ephedrine)
4. Fluoxetine

Digoxin: (Digitalis)

-Enhances myocardial contractility, ↓ HR, & slows impulse propagation through the AV node.
-Used to treat CHF & SVT
-Inhibits the Na^+-K^+ pump causing ↑ intracellular Ca^{++} accumulation.
-Work by decreasing Phase 4 depolarization of the SA node
-Hypokalemia, hypercalcemia, and hypomagnesemia increase the likelihood of digitalis toxicity.
-Hyperventilation should be avoided because it creates a relative hypokalemia. Hypokalemia causes binding of digitalis to myocardial cells, resulting in an excessive drug effect.
-Eliminated primarily by the kidneys, 35% daily

Three side effects of tricyclic antidepressants:

(Amitriptyline)
1. Anticholinergic effects (dry mouth, blurred vision, tachycardia)
2. Orthostatic hypotension
3. Sedation

Tricyclic antidepressants interact with:

1. Anticholinergics (atropine, scopolamine)
2. Sympathomimetics (ephedrine)
3. Inhaled anesthetics (↑ dysrhythmias)
4. Antihypertensives (rebound HTN)
5. Opioids (↑ analgesia & respiratory depression)

Anaphylactic Reaction: (Type I hypersensitivity reaction)

Antibody **Ig E** (immunoglobulin E) is produced in response to an antigen (foreign protein). Upon a second exposure to the antigen, Ig E on the surface of mast cells and basophils triggers the release of mediators including histamine. This causes bronchoconstriction, upper airway edema, vasodilation, increased capillary permeability, and urticaria. Life-threatening.

Anaphylactoid Reactions do not involve Ig E. Foreign substances (i.e. drugs, hetastarch) directly stimulate the emptying of mast cells and basophils.

Anaphylactic and Anaphylactoid reaction = same S&S

Top 5 Causative of Anaphylactic Reactions

NMB 60%
Latex 17%
Abx 15%
Colloids 4%
Hypnotics 3-4%

Latex Allergy

1.multiple surgical procedures
2. health care personnel
3 atopic people w/ hx of multiple allergies
4. food allergies: bananas, avocados, chestnuts, stone fruit

Coagulation:

Intrinsic Path = XII, XI, IX, VIII (12,11,9,8)
Extrinsic = III, VII (3,7)
Final CP = V, X, I, II, XIII (5,10,1,2,13)
Not in Liver: III, IV, VIII (3, 4, 8)
Vitamin K Dep: II, VII, IX, X (2,7,9,10)

Heparin = Intrinsic = aPTT & ACT
Coumadin = Extrinsic = PT & INR

Bleeding time	3-10 min. – Platelet function
PT	12-15 sec - extrinsic pathway
PTT	25-35 sec. – Intrinsic pathway
ACT	80-150 sec.
TT	9-11 sec. – final common pathway

One PRBC = ↑ Hct 3-4% 1g/dl
1cc/kg PRBC= ↑ Hct 1%
1 unit plts = ↑ 5,000-10,000 mm^3
Massive transfusion = 1 complete blood volume in 24 hours

PRBC

Universal Donor = O Universal recipient = AB

Platelets

Universal Donor= AB Universal Recipient=O

Coumadin drugs competitively inhibit vitamin K so synthesis of Vitamin K-dependent factors (II, VII, IX, and X) is diminished.

Cryoprecipitate is the fraction of plasma that precipitates when FFP is thawed at 4^0 C. (The drug of choice for Von Willebrand's disease)
Cryoprecipitate contains factors I, VIII, XIII

Aprotinin inhibits plasmin and therefore inhibits the breakdown of fibrin.

FFP

10-15mg/kg
Contains all clotting factors but plts
Uses: isolated coagulation factor deficiencies, reversal of Coumadin, liver dz- reverse coagulation issues, after massive transfusion and still bleeding

Max Allowable EBL:(≈20% EBV)

$$MABL = EBV \times \left[\frac{Hct - {}_{a}Hct}{Hct}\right]$$

Hct = 3 x Hbg

PRBC replacement: Hct of PRBC = 75

$$PRBC\ (ml) = \frac{(Blood\ loss - MABL) \times desired\ Hct}{Hct\ of\ PRBC}$$

Estimated Blood Volume (EBV):

Premie	(< term)	95 ml/kg
Term		90
Infant	(< 6 wks)	80
Toddler	(6 wks-2yrs)	75
Child	(2 yrs-12yrs)	72
Men		70-75
Women		65

Total Body Water (TBW):

Adult	60% = 42L
Neonate	80%
Premie	90%

ICF: 60-66% 25-28L
ECF: 33-40% 14-17L
Interstitial fluid 80%
Plasma water 20%

Fluid replacement:

4 ml/kg first 10 kg (1-10kg)
2 ml/kg second 10 kg (10-20)
1 ml/kg remainder (20+)
Short cut: Add 40 to your total Kg (if >20)

Fluids:

LR: pH 6.5 - contains K 4, Na 130,lactate 28– hypo (osm 273)
Too much = metabolic alkalosis
NS: Na = 154 meq/L = Isotonic (osmol = 308)
Too much= hyperchorlemic acidosis
D5: Hypotonic (osmolality 252 mOsm)
5% Albumin- colloid osmotic pressure of 20mmHg
Normosol- No Ca^{++} (osmol 294)
Hespan: 6% hydroxyethyl starch in NS =/> 20ml/kg/day = ↑ serum amalayse levels
Hextend- 6% hydroxyethy starch in solution w/ electrolytes, glucose and lactate

Isotonic: ~ 285 mOsm/L- effective osmolality close to body fluid
CELLS DO NOT SWELL OR SHRINK
Hypotonic ~ <285 mOsm/L -- ↑ vascular volume
Hypertonic ~ > 305 mOsm/L -- ↓ cells shrink

P_{osm}= 2 x plasma [Na^+]

Dextran 40:

Improves blood flow through the microcirculation presumably by decreased blood viscosity. Maximal dose = 20 ml/kg/day or about 58 ml/hr for 70 kg patient.
-Side effects of excessive Dextran 40 include: interference with blood typing, prolonged bleeding time, renal failure, and anaphylactoid reactions.

Five adverse effects of Mannitol administration:

1. Pulmonary edema and Cardiac decompensation
2. Rebound ↑ ICP
3. Hypovolemia
4. Hyperkalemia
5. Hyponatremia

Virchow Triangle

1. endothelial injury
2. stasis or turbulent blood flow
3. hypercoagulability of blood

Sickle Cell Disease:

A hereditary hemolytic anemia resulting from the formation of an abnormal hemoglobin (Hb S). Red cell survival is reduced to 10-15 days, compared with up to 120 days in normal individuals. Sickling occurs only under extreme hypoxemia or in low-flow states.
The goal of exchange transfusions is to decrease blood viscosity and achieve a Hct of 35-40% with 40-50% normal hemoglobin (Hg A_1). Avoid hypo- and hyperthermia, acidosis, and even mild degrees of hypoxemia, hypotension, or hypovolemia. Generous hydration and a relatively high (>50%) FiO_2 are desirable. Many avoid the use of tourniquets.

Acute chest syndrome – dyspnea, chest pain, hypoxemia

Normal hemoglobin (Hemoglobin A) has:

& { 2 beta globin chains
2 alpha globin chains

Sickle Cell Anemia is a mutation of the beta globin chains – glutamic acid instead of valine

Four diseases associated with thrombocytopenia:

1. Chemotherapy or unrecognized cancer
2. Liver disease and splenomegaly
3. DIC
4. Pre-eclampsia

Porphyria

Metabolic d/o affecting biosynthesis of heme = thick blood
Signs & Symptoms
1. Acute abdominal pain, N & V
2. Neurotoxicity: confusion, SIADH, difficulty swallowing, HTN & tachycardia
3. Sensory & motor neuropathies

AVOID Triggering Agents: KEPT MAN

Barbs	Benzos	Ketamine	Etomidate
Nifedipine	Ketorolac	Enflurane	Sulfamides
Phenytoin	Hydralazine	mepivicaine	lidocaine

GA- no regional

Hemophilia : x-lined recessive

A- Factor VIII Deficiency
B- Factor IX Deficiency

Prolonged PTT & normal PT
Tx: Factor Concentrates

Polycythemia:

Abnormal ↑ Hct — can develop AVWD
50-60% Hct = HA — Ideal Hct = 33-36
Tx: phlebotomy, avoid dehydration

Heat Loss:

Radiation 40%>Convection 32%>Evaporation 28%>Conduction

For each 1^0 C ↑ in body temperature,
Basal metabolic rate ↑ > 7%

The center for **Heat Loss** is located in the anterior (preoptic) hypothalamus and the **Heat Gain** center is located in the posterior hypothalamus.

Great decrease of core temp occurs in **1st hour** of surgery

Operating room temp #1 critical factor min OR temp 21C

Hypothermia

↑ SVR and myocardial depression (↓ CO)
↑ blood viscosity, left shift of oxyhemoglobin dissociation curve, impaired coagulation, and thrombocytopenia
↓ drug elimination 2^0 ↓ hepatic blood flow and metabolism.
↓ Renal blood flow and clearance slow renal excretion of drugs.

Eight Physiologic effects of Hypothermia:

1. ↓ O_2 consumption (↓ CO_2 production)
2. ↑ SVR
3. Cardiac dysrhythmias
4. Left shift oxyhemoglobin dissociation curve
5. Coagulopathy
6. ↑ Blood viscosity
7. Impaired renal function
8. ↓ Drug metabolism

Natriuretic Peptides
ANP- atrial – from atrial muscle in response to local wall stretch
BNP- brain- ventricle muscle when distended
CNP- endothelial walls
Urodilitation- lower urinary tract
-cGMP- mediates
-Induces vasodilatation of arterial and veins = ↑ RBF & GFR
-Suppress actions of NE, angiotensin, endothelin,
-Inhibit – renin, angiotensin II, aldosterone
↑ ANP & BNP = mortality predicator in CHF

Five second messengers:
1. cAMP
2. Calcium ions
3. Calmodulin
4. cGMP
5. Inositol Triphosphate (IP_3)

4 Reasons for difficulty breathing after Thyroidectomy:
1. Laryngeal edema
2. Bilateral cord paralysis
3. **Hematoma formation**
4. Hypocalcemia 2^0 hypoparathyroidism

Six hormones of the Anterior Pituitary: (Adenohypophysis)
(*Blood flow through Hypothalamic-Hypophyseal Portal System*)
1. Adrenocorticotropic hormone (ACTH)
2. Thyroid stimulating hormone (TSH)
3. Growth hormone (GH)
4. Prolactin
5. Leutinizing hormone (LH)
6. Follicle stimulating hormone (FSH)

2 hormones of the Posterior Pituitary: (Neurohyphosis)
1.Anti-Diuretic hormone (ADH)
2. Oxytocin

Thyroid:
Regulated by TSH release from anterior pituitary
93% Thyroxine (T4)
7% Tri-iodothyronine (T3)

About 80% of Tri-iodothyronine is produced outside the thyroid gland by de-iodinazation of thyroxine.

Tri-iodothyronine (T3) is four times more potent than thyroxine (T4).

T4 is converted to T3 in the tissues

Thyroglobulin = protein

Serum thyroxine levels < 1 mcg/dl → severe hypothyroidism and warrant delay of an elective case or treatment with thyroid hormone prior to emergency surgery.

Normal serum thyroxine levels is 5-12 mcg/dL

S/S of Grave's Disease: (Hyperthyroidism)
1. Intolerance to heat
2. ↑ Sweating
3. Mild to extreme weight loss
4. Varying degrees of diarrhea
5. Muscular weakness
6. Nervousness
7. Extreme fatigue
8. Inability to sleep
9. Tremor of the hand
10. Exothlamous

Beta antagonist is the drug of choice for treating hyperthyroid-related ventricular dysrhythmias.

Four causes of Hypothyroidism:
1. Subtotal lobectomy of thyroid
2. Goiter
3. Autoimmune disease (myxedema)
4. Radiation therapy of thyroid
Can cause cretism in infant = large tongue

Hypothyroidism:
S/S
1. Slow mental function
2. Slow movements (weight gain)
3. Dry skin
4. Cold intolerance
5. Depressed ventilatory responses
6. Abnormal cardiac conductivity (↓ cardiac function)
7. Renal disease

↓ Anesthesia requirements – however, no Δ in MAC

Thyroid Storm
Severe exacerbation of hyperthyroidism
S/S - 6-8 hrs. post-op
1. Hyperthermia 2. Tachycardia 3.CHF
4. Dehydration 5. Shock 6. Hyperglycemia
Tx: Na Iodide, cortisol, propranolol, Propylthiouracil
Thryotoxosis d/t sudden excessive release of thyroid gland horm

Difficulty breathing after thyroidectomy:
1.Laryngeal edema
2. Bilateral cord paralysis
3. *Hematoma formation #1*
4. Hypocalcemia 2nd to hypoparathyroidism
Parathyroid:
Regulates Ca^{++} & Phosphate

Preop parathyroid: correct volume, Avoid LR, ↑ NDMB

Parathyroid hormone ↑ $[Ca^{++}]$ by:
1. ↑ Absorption of Ca^{++} from intestine
2. ↑ Reabsorption of Ca^{++} from renal tubule
3. ↑ Resorption of Ca^{++} from bone

Complications of Parathyroidectomy:
1. Hypocalcemia
2. Parasthesias
3. Muscle spasm
4. Tetany
5. Laryngospasm
6. Bronchospasm
7. Apnea
8. Hematoma
9. Airway compromise
10. Pneumothorax

More excitable d/t Δ of threshold closer to RMP

Major postop concern after Parathyroidectomy:
(Airway obstruction)
Laryngospasm 2^0 hypocalcemia
Bilateral recurrent laryngeal nerve damage
Hematoma

Ca > 5.5 mEq/L
Ionized Ca^{++} >2.5 mEq/L

Osteitis Fibrosa Cystica is a bone disease caused by hyperparathyroidism. Leaking of Ca out of bones= broken and brittle bones

Calcitonin promotes the deposition of calcium in the bones and decreases $[Ca^{++}]$ in the ECF. (opposite of PTH)

Clinically significant hypocalcemia:
ECG changes (prolonged Q-T interval)
↓ Myocardial contractility

S/S of hypocalcemia following parathyroidectomy:
1. Perioral parasthesias
2. Restlessness
3. Neuromuscular irritability
Chvostek's sign
Trousseau's sign
Inspiratory stridor

Five complications of RRP:
1. Hemorrhage
2. Compromised ventilatory function
3. Air embolism
4. Damage to obturator nerve 2^0 retractors
5. DVT

Four effects of acidosis on CNS function:
1. Depressed neuronal activity (coma)
2. Cerebral vasodilation (↑ CBF, ↑ ICP)
3. ↓ Cerebral perfusion pressure (cerebral ischemia)
4. ↑ Seizure threshold

Anion Gap: = $[Na^+] - [Cl^-] - [HCO_3^-]$
Normal range = 9-15 mEq/liter
Used in the differential diagnosis of metabolic acidosis.

Four causes of metabolic acidosis:
1. Ketoacidosis
2. Lactic acidosis
3. Renal failure
4. Toxic dose of salicylates
Metabolic Alkalosis
1. Vomiting
2. NG suctioning
3. Hypokalemia 2^0 diuretics

The kidneys excrete H^+ as titratable acids H_2PO_4 & NH_4^+
Plasma K^+ ↑ approximately 0.6 mEq/L for each 0.1 ↓ in pH

Six physiologic functions that require Ca^{++}:
1. Action potentials in smooth and cardiac muscle
2. Blood coagulation
3. Bone formation
4. Muscle contraction
5. Membrane excitability (Ca^{++} controls threshold)
6. Neurotransmitter release- Ca is REQUIRED

A rapid ↓ in plasma $[Ca^{++}]$ leads to skeletal muscle spasm (laryngospasm) and tetany.

Nine S/S of Hypocalcemia:
1. Numbness } earliest d/t neuronal membrane
2. Circumoral paresthesia } irritability
3. Confusion
4. Seizures
5. Hypotension
6. ↑ LV filling pressures (due to ↓ contractility)
7. Prolonged QT interval
8. Skeletal muscle weakness
9. Fatigue

Chvostek's sign (contracture of facial muscle with tapping) monitors hypocalcemia. d/t ↑ membrane excitability, closer to RMP

ECG changes with Hypocalcemia:
1. Prolonged QT interval
2. ↑ ST segment duration
3. Flat or inverted T-waves

Functions of Magnesium:
(It both resembles and antagonizes Ca^{++})
1. Functions as a cofactor in many enzyme pathways
2. Regulates the Na^+/K^+ pump
3. Regulates adenylate cyclase
4. Regulates slow Ca^{++} channels
5. It antagonizes Ca^{++} (an endogenous Ca^{++} channel blocker) = vasodilatation
6. Controls the threshold potential (membrane stabilizer)
7. Regulation of the release of acetylcholine from nerve terminals

Hypermagnesemia:
Signs & Symptoms
1. Coma
2. Hypoventilation
3. Hypotension

↑ Mg = ↓ excitability

Caused from: excess dietary intake of Mg, excess ingestion of oral antacids, hypothyroidism, hyperparathyroidism, Addison's disease, & lithium therapy

Tx: forced diuresis with saline and loop diuretics

Hyperphosphatemia
Impairs coagulation by causing platelet dysfunction
Impairs ventricular contractility & leukocyte function

Causes of hypophosphatemia:
1. Ingestion of large # antacids containing aluminum & Mg
2. Severe burns
3. DKA
4. ETOH WD
5. Prolonged respiratory alkalosis
Tx: aluminum based antacids, Carafate, Ca citrate, dialysis

Eight signs of Hyponatremia: $[Na^+]$ < 135 mEq/L
(Normal $[Na^+]$ = 138-146 mEq/L)
1. Arrhythmias 2. Hypotension
3. Pulmonary edema 4. Mental changes
5. Muscle cramps 6. Weakness
7. Myoclonia 8. Edema
S/S develop with $[Na^+]$ < 120 mEq/L
Treatment is with diuretics and hypertonic saline
Hypernatremia = dehydration

Three factors that promote Hypokalemia:
(Normal extracellular $[K^+]$ = 3.8-5.0 mEq/L)
1. Alkalosis
2. Insulin
3. Beta-2 adrenergic stimulation } stimulates Na-K pump**3Na:2K**

Seven ways to treat Hyperkalemia:
1. Give calcium gluconate
2. Give glucose/insulin
3. Give sodium bicarbonate
4. Give diuretics (to increase excretion)
5. Give kayexalate (potassium exchange resins)
6. Use hemodialysis
7. Hyperventilate the patient

Plasma [K^+] is ↓ 0.5 mEq/L for each 10 mmHg ↓ in $PaCO_2$

Cardioplegia – 15-40 mEq/L of K^+

Calcium should be given to the hyperkalemic pt. when ventricular dysrhythmias appear. (↑ threshold away from RMP)

Medications that cause an ↑ K
Triamterene, spironolactone, NSAIDs, ACE inhibitors, BB

ECG Δ with Hyperkalemia
Prolongation of P-R interval
Widening of QRS
Peaked or tented T waves

ECG Δ with Hypokalemia
Prolongation of P-R & Q-T interval
Flattening of T waves
Appearance of prominent U wave

Addison's disease: glucocorticoid & mineralocorticoid deficient
Hypoadrenocorticism- autoimmune destruct of the adrenal cortex
S/S
1. Hypotension
2. Hyponatremia } ↓ Aldosterone
3. Hyperkalemia
4. Hypoglycemia (2^0 ↓ cortisol levels)
5. Hemoconcentration (d/t to ↑ H_2O excretion 2nd to hyponatremia)
6. Skin pigmentation

Give glucocorticoid intraop

Cushing's disease:
Hyperadrenocorticism- ↑ cortisol & ↑ ATCH in anterior pituitary
1. HTN (2nd to Na retention)
2. Hypokalemia
3. Hyperglycemia (2^0 ↑ cortisol levels)
4. Moon face & buffalo hump
5. Skeletal muscle weakness
6. Skin pigmentation

Conn's Disease:
Primary hyperadolteronism
↑ Na^+
↓ K^+

Pheochromocytoma:
Tumor of the adrenal medulla chromaffin tissues which results in an ↑ catecholamine release
S/S:
1. Paroxysmal HTN
2. Diaphoresis
3. Tachycardia
4. Headache

Tx:
α block- phenoxybenzamine 20-30mg/day & ↑ to 60-250mg/day,
β block- tx for tachy
Avoid: Trimethaphan, droperidal, histamine

Multiple Endocrine Neoplasia: (MEN)
A group of syndromes characterized by tumor formation in several endocrine organs.
MEN I = tumors in pancreas, pituitary gland, & parathyroid gland
MEN II = medullary thyroid carcinoma, pheochromocytoma, and hyperparathyroidism (type IIa) or multiple mucosal neuromas (type IIb or type III)

Kidney: 25-30% CO
Functions:
1.ECF composition: Osmolality: 285-305 mOsm/kg
Urine osmolality 50-1200 mOsm/kg H_2O
2. Maintenance of EFV- NA & H_2O excretion
3. Endocrine:
Erythropoietin- CRF = anemia
RAA System- BP reg, Na/K excretion
Vitamin D: CRF = hypocalcemia
4. Regulation of arterial BP

RBF = (MAP- Venous Pressure) x Vascular resistance
Autoregulation – MAP 75-160 mmHg
Filtration ceases MAP <60mmHg

Labs: Creatine Clearance is single best indicator of renal status
Creatine: 0.7-1.5 mg/dl BUN: 10-20mg/dl
GFR: nml 95-150ml/min, mild dsyfx 50-80, mod dsyfx <25, failure <10 ml/min

Nephron Sections:
1. Glomerulus: freely filters Na, Cl, K & H_2O
2. Proximal tubule: 67% of glomerular filtrate- reabsorbed H_2O > Cl > Na = K, **ONLY** place permeable to **glucose**
Filtration, reabsorption and secretion
3. Descending LOH: Urea & H_2O (no Na, Cl or K)
Osmotic gradient via countercurrent multiplier
4. Ascending LOH: Filters Na, Cl, K – No H_2O in thick branch
Loop Diuretics – Na^+, K^+, $2CL^-$ inhibit reabsorption
SE: ↓ K, fluid volume deficit, orthostatic HoTN, reversible deafness (CNVIII)
Lasix ↑ prostaglandins= venodilitation = ↓BP
5. Distal Tubule: Filters Na & Cl (No K or H_2O)
Early: Thiazides: ↓ K
Late & CC: Potassium Sparing Diuretic: ↑ K
Triamterene, amiloride Spiratalctone
ADH & Aldosterone
6.CollectingDuct:Na & Cl-out, K-In, Low perm H_2O, (no urea)
Site of Action ADH & Aldosterone (principle cells)

Cortex: Glomeruli, Proximal Tubules, Distal Tubules
Medulla: loops of Henle, Collecting Ducts

*Inner stripe of the outer medulla= most vulnerable to ischemia

Carbonic Anhydrase Inhibitor:
Acetazolamide(Diamox)-↓ IOP 2nd to ↓ aqueous humor formation
Inhibit carbonic anhydrase in proximal tubule = ↓ Na reabsorption = diuresis – Hyperchorlemic Metabolic Acidosis

Mannitol:
↑ Osmotic pressure in renal tubule= ↓ reabsorption of H_2O
S/E: ↓ K^+

Renal Failure:
S/S & lab findings normal until 40% ↓ in Fx

	FE_{NA}	Urine Sodium (mEq/L)
Prerenal Failure	<0.01 (1%)	< 20
Renal Failure	> 0.03 (3%)	>40

Anuric: UO <100ml/day or < 0.5ml/kg/hr
Oliguric: UO 100-400 ml/day
NonOliguric: UO >400ml/day

Electrolytes: ↑ K, Mg, Phos, ↓Ca , Na , metabolic acidosis

Infection is # 1 issue and cause of death

Fenoldapam DA1 agonist: ↑ RBF (6x more potent than dop)

Avoid Meperidine & Morphine (metabolite-6)

ADH: (vasopressin) : *NA^+ Concentration*
Synthesized Periventricular and Supraoptic nuclei of the hypothalamus, stored in and released from the posterior pituitary (neurohyposis).
1. 1^0 on collecting ducts in the nephron = passive water reabsorption
2. Vascular smooth muscle and cardiac myocytes.
3.Dramatic vasoconstriction and has an inotropic effect

Released in response to ↑ ECF **osmolarity**, ↓ plasma volume, & drugs, stress, H_oTN, Pain, CPAP, PEEP, VA, Trauma
Works in the distal tubule & collecting duct to ↑ H_2O reabsorption, In the absence of ADH the collecting duct and distal tubule are impermeable to H_2O ie: excretion happens.

ADH = Osmolality = ↑ [Na^+] (*concentration*)

Extra Cellular Fluid Volume: Na^+ = ECFV

1. 1^0 on collecting ducts in the nephron = passive water reabsorption
2. Vascular smooth muscle and cardiac myocytes.
3.Dramatic vasoconstriction and has an inotropic effect

Aldosterone: *VOLUME*
It controls *intravascular **volume*** and works in the **collecting duct** and the late distal convoluted tubule.
Increases Na^+ reabsorption (conserved), and K^+ secretion (excretion).
Release is triggered by **Angiotensin II and ↑ [K^+]**; also ↑ ACTH and low serum Na^+. It also regulates the total amount of Na^+.
Aldosterone (a mineralocorticoid) is secreted by the zona glomerulosa of the adrenal cortex.
Cortisol (a glucocorticoid) is secreted by the zona fasciculata, the middle zone of the adrenal cortex.
Epinephrine (80%), norepinephrine (20%), and dopamine are catecholamines released from the adrenal medulla.
This is controlled by sympathetic preganglionic neurons that release acetylcholine to nicotinic receptors.

3 Stimuli for the release of Renin:
1. ↓ **Renal perfusion pressure** *Normal GFR* = 125 ml/min
2. Hyponatremia
3. Sympathetic NS stimulation of beta-receptors in JG apparatus

Clinical features of Diabetes Insipidus:
Polyuria, > 2-15 L/day- HALLMARK
Hypernatremia (plasma hyperosmolality)
Dilute urine (osmolality < 200 mOsm/kg)
Neurogenic vs. Nephrogenic – give ADH to Dx

S/S of Diabetes Insipidus:
1. Polydipsia (↑ thirst)
2. **Polyuria –** (2-15L/Day) **HALLMARK**
3. Hypernatremia
4. Hypovolemia
5. Hypotension
6. ↓ Urine osmol (<200 mosm/kg)
7. ↓ specific gravity (1.005 or less)

Diagnosis of SIADH: ↑ ADH (intracranial tumors)
Decreased plasma osmolality (< 270 mOsm/kg)
↑ urine [Na^+] (> 100-150 mOsm/kg)
Hyponatremia d/t retention of H_2O
Tx: remove underlying cause, Limit H_2O , 3% Saline, demecolcine

Liver: 25% CO
Blood Flow:
70% Portal vein - splenic & mesenteric } 50:50
30% Hepatic artery } Oxygen supply

"Arterial Buffer Response"
An increase in hepatic arterial flow in response to a decrease in portal blood flow (1) to maintain hepatic O_2 supply, and (2) to maintain total hepatic blood flow, which is essential for clearance of many compounds.

Hepatic Blood Flow
Arteries: α-1, β-2, D1 &cholinergic =blood flow receptors: Ohms
30% of Blood flow, 50% of oxygen supply
Veins: α-1 & D1- blood flow depend on GI & Spleen
70% of blood flow, 50% of oxygen supply

Seven functions of the Liver:
1. Storage and filtration of blood (500ml)
2. Metabolic functions such as carbo, fat, and protein metabolism
3. Secretion of bile
4. Storage of vitamins
5. Blood coagulation
6. Storage of iron
7. Detoxification & excretion of drugs

P450's
Inducers: chronic ETOH, barbs, ketamine, benzo's, phenytoin, cigarette smoke, St. Johns wort

Inhibition: cimetidine, chloramphemocel, fluoxetine, grape fruit
Bile vomitus is usually alkaline. This is because large amounts of alkaline phosphatase are normally excreted in the bile. A three-fold or greater increase in alkaline phosphatase is indicative of biliary tract obstruction.
Puke your acids = alkalosis
Poop your bases = acidosis

Alcoholic Hepatic Issues/Liver Disease
Thrombocytopenia, leukopenia, anemia, megoblastic anemia d/t folate deficiency
↑ MAC
DT's 24-96 (1-4 days)
Maintain BP- ETOH depends on ARTIERAL FLOW
Isoflurane- agent of choice
↑ Vd= ↑ loading dose, smaller main = prolonged Vec, Roc, Pan
Labs: ↑ total bilirubin, ↑AST/ALT (ALT more sensitive), ↓ albumin & PT
Hemodynamics: ↓SVR, ↑CO, ↑circ blood volume, ↓portal blood flow, ↓ plasma osmotic pressure

Increased surgical risk in Liver Failure patient
1. High bilirubin >3.0 mg/dl
2. Low albumin < 3.0 mg/dl
3. Severe ascites
4. Coma 5. Poor nutrition

Mendelsons Syndrome: (aspiration pneumonitis)
Gastric pH < 2.5
Gastric volume <25 ml

Seven conditions that delay gastric emptying:
1. Obesity
2. Pregnancy
3. Opiods
4. Diabetes mellitus
5. Trauma
6. Pain
7. Anxiety

Drugs that ↓ LES Tone: (↑ risk of aspiration)
1. Anticholinergics,
2. Dopamine
3. Thiopental
4. Opioids,
5. Propofol,
6. Tricyclic antidepressants
7. Sodium nitroprusside.

Patients who benefit from antiemetics:
1. Eye surgery patients
2. Gynecological patients
3. Obese patients
4. History of vomiting
5. ECSW Lithotripsy

Antiemetics:
Competitive Dopamine Antagonists;
Droperidal, Compazine, Metoclopramide
Do not give to Parkinson's Patients

Droperidol:
1. An antiemetic (2^0 dopamine blockade)
2. Produces extrapyramidal signs (2^0 dopamine blockade)
3. May ↓ BP (2^0 weak alpha blockade)

Treatment for droperidol-induced extrapyramidal symptoms is an anticholinergic (Benadryl or benztropine, *Cogentin*).

Do not give to: Pheo or Parkinsons

Metoclopramide: **(NO Δ in pH)**
1. Relaxes pyloric sphincter
2. Promotes gastric motility = ↑ gastric emptying
3. ↑ Lower esophageal sphincter tone = ↓ reflux
4. Relaxes the small intestine
5. Has antiemetic actions

It also inhibits plasma cholinesterase

Used to treat:
Diabetic gastroparesis
Reflux esophagitis
Pts. at risk for aspiration pneumonitis (Mendelsons syndrome)
Parturients

Drugs that ↑ gastric pH and ↓ gastric volume: (H_2 blockers)
1. Cimetidine (Tagamet) **inhibits P-450 the most
2. Ranitidine (Zantac) ** most potent
3. Famotidine (Pepcid)

Ranitidine ↓ gastric volume & ↑ pH the most

Inhibits the cytochrome P-450 system.
Cimetidine>Ranitidine>Famotidine=0

Drugs that interact with Cimetidine:

1. Propranolol	2. Labetalol
3. Diazepam	4. Halothane
5. Lidocaine	6. Phenytoin
7. Coumadin	8. Aminophylline
9. Meperidine	10. Quinidine
11. Tricyclic antidepressants	12. Ca^{++} channel blockers

Proton Pump Inhibitors:
Omeprazole- inhibits the hydrogen pump - = ↓ secretion of HCl into the lumen of stomach
*most powerful agent for ↓ acid secretion

Antacids:
Raise gastric pH by neutralizing HCl also ↑ gastric volume
Immediate Effect
Non-Particulate – better for aspiration risk; however, loose effectiveness after 30-60 mins

Chemoreceptor Zone:
4th ventricle (area posterema)

Neurotransmitters & Receptors in N/V:
Dopamine & DA-2
Serotonin & 5-HT
Acetylcholine & muscarinic
Substance P & NK-1

Carcinoid Syndrome
S/S caused by vasoactive substances released from enterochromaffin tumors or carcinoid tumors
Most carcinoid tumors are in GI tract- especially appendix
Cause overproduction: **serotonin**-**hallmark**, bradykinin, histamine, kallikrein, prostaglandins

**Avoid activating tumor: HoTN, catecholamine release, histamine releasing medications
*HoTN- volume expanders
*right sided ♥ failure

Five clinical manifestations of nonintestional tumors:
1. Cutaneous flushing
2. Bronchospasm
3. Diarrhea
4. Large BP swings
5. Supraventricular dysrhythmias
**you don't see this with intestinal tumors d/t first pass metabolism in the liver

Octreotide (somatostatin) used to blunt vasoactive and bronchoconstritive effects of carcinoid products

Pancreas: Endocrine
Beta cells: insulin
Alpha cells: glucagon
Delta cells: somatostatin

Evaluation of pancreatitis:
1. Malnutrition
2. Abnormal liver FX
3. ETOH WD (alcoholism frequent cause of pancreatitis)

Symptoms of pancreatitis:
1. Dehydration
2. Hypocalcemia
3. Hyperglycemia
4. ARDS
Alcoholism is a frequent cause of pancreatitis.

5 "F's" of Cholelathiasis
1. Fat 2. Female 3. Fertile 4. Forty 5. Fair

Insulin is not needed by brain or RBC to utilize glucose

Insulinoma:
Insulin secreting tumor causing massive insulin release.
Tx: tumor resection- Childs Procedure
Hypoglycemia- during resection
Hyperglycemia- after resection
Check Blood sugar every 15 mins

Diabetic Neuropathy
40-50% of IDDM
Anesthetic considerations: orthostatic Hotn, Silent MI, Gastroparesis
HR response blunted w/ use of antimuscurinics - Atropine

Hypoglycemia
S/S: Diaphoresis, HTN, Hunger, feeling of impending doom

A1C = 6%

DM with autonomic neuropathy 65% insulin dependent
3 Anesthetic concerns
Orthostatic Hotn, silent MI, Gastroparesis

6 Cardiac Concerns
Orthostatic Hotn, Resting tachycardia, absence of beat to beat variability in HR, dysrhythmias, silent MI, painless myocardial ischemia

HR response blunted in these medications:
Antimuscarinics & Beta-Blockers

MOA of DM neuropathy: High concentrations of glucose are converted to sorbitol within nerve via the polyol pathway. The high levels of sorbitol (and fructose) that develop ↓ ATP→ ↓ NO levels. Because nitric oxide acts as a vasodilator, the loss of nitric oxide leads to unopposed vasoconstriction and ischemia of the nerve

Thymus
Where T-Lymphocytes mature
Thymus is most commonly removed d/t Thyoma & MG
Pneumo most likely on **RIGHT** side
Thymic vein drains into innominate vein

Scoliosis
Lateral curvature of the spine
Restrictive lung dz- ↓ VC, TLC, IV, FRC, IC, ERV
FEV_1/FVC = normal Deliberate Hotn MAP 60-65
Mitral valve prolapse Anticipate blood loss

Muscular Dystrophy
Painless degenerative & atrophy of skeletal muscles
Duchene's (most common & severe)- x-linked, recessive
EKG- prolonged PR, QRS, ST abnormalities, BBB, Q waves, R waves (tall), CHF
Resp: ↓TLC & RV, recurrent pulm infections
NO succs (↑MH)

Multiple Sclerosis- MS
Random & multiple sites of demyelination of corticospinal tracts in brain & SC (NO PERIPHERY)
No spinals (may exacerbate)
Avoid – rises in temps
Hyperkalemia in response to Succs

Parkinson Disease
Loss of dopanergic fibers basal ganglia (dop is a inhibitory NT)
s/s: skeletal muscle rigidity, resting tremor, diaphragmatic spasm
Tx: Levodopa
Levadopa S/E: dyskinesias, hallucinations, paranoia, and mania. ↑ in cardiac contractility & HR. Orthostatic HoTN & N/V
Avoid: droperidal, reglan, Compazine

Rheumatoid Arthritis
Chronic inflammation disease w/cervical vertebral involvement
Restrictive lung disease, pulmonary fibrosis, FOB, pulm fx test
Hoarseness or stridor – Cricoarytenoid involvement
♥ Δ's- pericardial effusion, aortic regurg, cardiac valve fibrosis, anemia,

Cyanide Toxicity: Cyanide binds to cytochrome oxidase resulting in inhibition of oxidative phosphorylation causing inhibition of cell respiration. (Nitroprusside)

Myasthenia Gravis- MG
Chronic autoimmune @ NMJ, weakness
Ptosis & diplopia – most common initial s/s
Cardiomyopathy & hypothyroidism
Tx: steroids, anticholinesterase meds, plasma phoresis
Surgery: Thymectomy
Avoid muscle relaxants

Lambert-Eaton Syndrome ≈ Myasthenic Syndrome:
A disorder of neuromuscular transmission associated with carcinomas. An autoimmune disease in which **immunoglobulin G (IgG)** antibodies against voltage-gated sodium channels cross-react with calcium channels at the neuromuscular junction. The result is a decreased release of acetylcholine in response to nerve stimulation. Increased sensitivity to NDMRs and succinylcholine is seen.

↓ Weakness with ↑ activity

Ankylosing Spondylitis:
A familial d/o associated w/ histocompatibilityantigen HLA-B27.
Low back pain associated w/ early morning stiffness young men progressively restricted movement of the spine.
Some patients develop arthritis of the hips and shoulders.
♥ issues **Stress steroid dose if needed**

Lupus
characteristic manifestations: antinuclear antibodies, nephritis, serositis, thrombocytopenia, or a characteristic rash.
S/S: symmetric arthralgias hands, wrists, knees, and ankles, doesn't include spine. 33% - Cognitive dsyfx

Complications of TURB:

1.Blood Loss	2. Hypothermia
3. bactermia	4. Bladder perforation

Peritoneal Invasion – Sign and symptoms

Awake Patient	**General Anesthesia**
Sudden, severe, abd pain	Unexplained HTN (HoTN rare)
Should pain	Tachycardia
NV	
Abdominal rigidity	

TURP
Preference for SAB: ↓ blood loss, ↓ thrombus risk, atonic bladder, prevents postop bladder spasms, awake pt. can assist detecting issues

Complications:

1.Blood Loss	2. Venous absorption of irr fluid
3. venous sinuses of prostate	4.↑ intravascular volume
5. Dilutional hyponatremia	6. Bactermia
7. hypothermia	8. Coagulopathy

Nonelectrolyte Solutions for TURP
hypotonic/nonelectrolyte
Glycine 1.5%–↑glycemia = temp blindness, ammonia tox, N/V, HA, ECG Δ,
Sorbitol- 2.7%
Mannitol- 0.54%

Fluids can cause bladder distension & toxicity

Bladder Perforation during TURP –Signs & Symptoms:
1.Abdominal pain & spasm
Suprapubic fullness
HTN & tachy—followed by sudden & severe HoTN

TURP Syndrome
Triad of S/S: ↑ PP, bradycardia, mental status Δ
Other symptoms: HTN, CV collapse, ↑ CVP, Dyspnea, N, anxiety

Tx: Give O2, notifiy surgeon, invasive monitors, blood to lab, 12-lead ECG,

Central Pontine Myelinolysis - complication from TURP syndrome- d/t the rapid tx of hyponatremia

Management: stop resection, Labs, fluid restriction, Lasic, NS3%
Amount of NS 3% = Dose (mEq) = kg x (140-(Na)mEq/L)
Stop NS 3% when NA > 120
Na^+ Levels:
120 mEq/L = restlessness & **confusion**, (psbl. wide QRS), N/V
115 mEq/L = nausea, somnolence, **ECG Δ's** (wide QRS, ST elevation)
110 mEq/L = seizures & coma with VT, VF

****Irrigating fluid = 20ml/min of surgery time**
****Blood loss is ~ 3-5ml/min of resection time**

Gas Embolism – laparoscopic surgery
S/S: HoTN, Tachy, bilateral wheezing & ↓ $ETCO_2$
Halt insufflation, eliminate N_2O, release pneumoperitoneum, place in ***LEFT*** *lateral decub*, aspirate gas via central venous cath

Signs of Fat Embolism during Surgery:
1. ↓ $ETCO_2$
2. ↓ PaO_2
3. ↑ PA pressure
4. ↑ $PaCO_2$
5. ↑ETN_2
6. Late signs: ↓ BP, ↑ HR, cardiac dysthymias, cyanosis, mill wheel murmur

Fat Emboli Syndrome – 12-72 hrs
4 major signs: axillary, subconjuctival petechial, hypoxemia, ↓CNS, pulm edema
5 minor signs: HR >100, pyrexia, retinal fat emboli, jaundice, renal Δ
Lab features: fat microglobumia (required), ↑serum lipase, anemia, thrombocytopenia, ↑ ESR
Triad: hypoxemia, mental confusion, petechial
Prophylaxis: corticosteroids – limiting endothelial damage by FFA
Tx: supportive

S/S of Intraop PE:
Sudden acute dyspnea!
1. Hypotension
2. Tachycardia
3. Hypoxemia, ↓ SpO_2
4. Bronchospasm
5. Hypocapnia, **↓ $ETCO_2$-** 1st sign d/t dead space ventilation
6. PVR > 300
7. + D-dimer

Rhabomylosis
Skeletal muscle necrosis d/t tissue injury
Myoglobin: oxygen transport protein
Common causes: major crush injury, thermal or electrical injury, arterial occlusion, acute muscle injury, prolonged immobility, compartment syndrome, MH, extreme lithotomy, hyperlordotic position

Surgical Lumbar Sympthathectomy
To improve blood flow to ischemic areas: causalgia, Raynaud's, frostbite, gangrene, ischemic ulceration of lower extremities
L_2-L_3
Cyanide Toxicity
Metabolic Acidosis
Cardiac arrhythmias
↑venous O_2 content
Tachyphylaxis
Tx: Na Thiosulfate 150ml/kg over 15 mins
Myocyte = muscle cell
Complex Regional Syndrome
Type 1 & Type II (causalgial- previous nerve injury documented)
Tx: physical therapy, gabapentin, tricyclics

Cor Pulmonale
ECG sings of cor pulmonale = peaked P waves in leads II, III, and aVF which are consistent with right atrial hypertrophy & right axis deviation and right bundle branch block which are consistent with right ventricular hypertrophy.

Surgical Stimulation
Intubation > upper ab surgery > breast surgery = lower ab = skin incision > skin closure

ASA classes
1-Healthy
2- mild systemic disease- HTN, DM, Anemia, age, obesity, chronic bronchitis
3- severe systemic disease- Cardiac dz with limiting activity, Uncontrolled HTN, DM w/vascular issues, MI or Angina
4- incapacitating systemic dz – CHF, persistent angina, advanced kidney or liver dz
5- moribund pt. – Not expected to live 24 hours after surgery- PE, AAA, cerebral trauma
6- brain dead- organ procurement
E- not elective- i.e appy

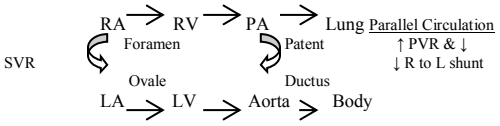
Mallampati
Class 1 – **P**illars, soft palate, fauces, uvula,
Class II- **U**vula, soft palate, fauces
Class III- **S**oft palate, uvula
Class IV- **H**ard Palate

ECT
1st – parasympathetic- HoTN, ↓HR **2nd**- sympathetic- HTN, ↑HR ↑CBF = ↑ICP Medication of choice: Brevital (0.5mg/kg)
Desired Duration = 30-60 seconds, sz sec needed= 400-700
Absolute contraindications:
1. Pheo
2. Recent MI (4-6 wks)
3. CVA <3mos
4. recent intracranial surgery- <3mos
5. Intracranial mass lesion
6. Unstable C-spine

Relative contraindications: angina, CHF, Pacer/ICD, pulm dz, major bone fx, glaucoma, retinal detachment, thrombophlebitis, pregnancy

Anesthetic goals: partial NMB to prevent long bone fx & unconsciousness

Fetal Circulation:



2 arteries-deoxygenation blood & 1 vein- oxygenated
Hemoglobin levels are lowest at 3 months of age 10-11mg/dl
Basal metabolic rate is highest at 2 years of age
Oxyhemoglobin curve: Newborn= Left shift, Right @ 3-4 mos.
Inspiratory pressure more negative than 25-40 to open alveoli
RDS-hyaline membrane disease, 50-75% mortality rate d/t result of inadequate surfactant in the alveoli

Premie: less than 37 weeks
Neonate: less than 30 days of age
Infant: 1-12 mos. of age
Children: 1-12 years of age
Post conceptual Age = gestational age + post maternal age

Apgar scores: 1 min = survival & 5 min = neurologic outcome

Differences in body systems in Pediatrics:
Cardiac:
CO of NN & infants is dependent on HR 2nd fixed SV & noncompliant & poorly developed L vent, depends on Ca^{++}
Innervation: Sympathetic = SPARSE, parasympathetic-complete

Respiratory:
↑ RR, ↓ Lung compliance + ↑ chest wall compliance = ↓ FRC (↑ IA induction), Deficient in Type I fibers, high closing volumes
Anterior and cephalad larynx (C3-C4), Adult C4-C5
Obligate nasal breathers
Cricoid cartilage narrowest point in children less than 5 yrs.
MAC highest 6 mos. of age

Kidney:
Incomplete glomerular development, ↓ size of glomerulus ↓ perfusion pressure, ↓ ability to concentrate urine, obligate Na losers

Hepatic:
Immature hepatic biotransformation, ↓ protein binding, fetal liver = hematopoiesis, ↑ risk of hypoglycemia, ↓ LBF

Neuro:	**Spinal Cord**	**Dural Sac**
NN	L3	S3
>1 y.o	L1	S1

Motor cortex- unmylenated and poorly developed until 2 years
Sensory cortex- mylenated @ birth, nerves poor develop until 3 mos.
CBF 40ml/100g/min (adults 100/100/g/min)
Neuro surgery: $PaCO_2$ → 20-25 mmHg

Thermo:
Non-shivering thermogenesis (inhibited by: IA, BB, fent, prop)
Best way to maintain infants body heat = heat up OR to 26C (78.8)
O_2 consumption 2x an adults
Other:
Large volume of distribution 2nd to higher total body water content
Higher ratio of body surface area to body weight

Cardiac Variables:
MAP = Gestational Age
Hypotension: SBP **NB** < 60 **1 year** < 70 **> 1yr** 70 + (age x 2)

ETT:

Diameter: 4 + Age /4 (cuffed)	Premie: 2.5,	Infant: 3.0,	3-12 mos: 3.5	
3 + Age/3 (uncuffed)		3.0	3.5	4.0
Length: 10 + age/2		10	11	12

Wt. in Kg	**LMA**	**Cuff vol**	**ETT**	**FOB**
NN- 5	1	4 cc	3.5	2.7
5-10	1.5	7	4.0	
10- 20	2	10	4.5	3.5
20-30	2.5	14	5.0	4.0
> 30	3	20	6.0	5.0

French Suction Catheter:
Neonate: 8 Fr, 2mos-2yrs: 10 Fr, 2-12 yrs: 14 Fr

Precordial Stethoscope: 3rd-4th intercostal left of sternal border

Medications:

Propofol	**Midazolam**
< 2 y.o = IV induction 3-4 mg/kg.	IV dose = 0.05mg/kg
>2 y.o. = 2.5-3 mg/kg	PO dose= 0.5 to 0.75 mg/kg
200-300mg/kg/min GA	

Flumazenil	**Thiopental**
IV 0.05 mg/kg → 1 mg total	4-6mg/kg
Succinylcholine = ↑ intubate dose	**NDMR**
IV 2-3mg/kg	Same dose as adults
IM 4mg/kg	
Myalgia, myoglobinemia, ↓ HR	

Prematurity:
Less than 60 weeks post conceptual = > risk of post anesthesia comp
Elective surgery: at least 60 weeks post conception
ROP- retinopathy of prematurity: vasoconstriction of retinal vessels- leading to permanent scarring, blindness and retinal detach
< 44 weeks (post conception)
Causes: Hyperoxia, hypercarbia, Hotn, Sepsis
Maintain PaO_2 60-80mmHg O_2 sats 89-94%,

Breathing Patterns:
Periodic breathing
Recurrent pauses in ventilation no longer 5-10 seconds. During REM sleep not associated with any physiologic disorder.
Central Apnea
Unexplained cessation of breathing > 15 sec or shorter resp pauses w/ HR < 100, cyanosis, pallor or loss of muscle tone. Premies.
Most important risk factor of postoperative apnea

Congenital Heart Issues:

	Foramen Ovale	**Ductus arteriosus**
MOA	1.↓ PVR & ↑ pulm flow 2. ↑ SVR = ↑ L vent pressure = ↑LAP	↑ PaO_2 & ↓ prostaglandins ↑ SVR & ↓ PVR
Functional	soon after birth	2-4 days
Anatomical	2-3 mos.	1-4 mos.
Shunt	Right to Left	Left to Right
MOA	RAP > LAP	RAP > LAP
Problem	Cyanosis	↑ work of left side of ♥ → Left ventricular hyertrophy
IV induct	Faster	Slower
Inhal induc	Slower	Faster

RAP > LAP (causes of return or continuation of fetal circulation)
1. Hypoxia
2. Hypercarbia
3. Acidosis
4. Hypothermic
5. Coughing, bucking & Valsalva
↑ PVR or ↓ SVR

ETCO2 underestimates AaCO2

Preductal Monitoring : Right Hand or Finger
Postductal: Left foot or toe
ABG's best obtained from Right Artery

Neonatal Coarctation of the Aorta
Narrowing of the descending aorta. If severe – perfusion is dependent on open PDA shunt. PGE1 use to maintain patency
BP monitoring in Right Radial artery

Congenital Diaphragmatic Hernia (CDH)
Foramen of Bochdaleck or anterior foreman of Morgagni
Larger on left & 90% of diaphragmatic hernias on left.
Maintain preductal saturation > 85 w/ PIP < 25 cm H_2O
Keep them breathing
Decompress stomach and avoid barotrauma
R to L shunt – monitor pre and post ductal perfusion
Cyanosis, Dyspnea & Dextrocardia

Tetralogy of Fallot
1. Large single ventricular septal defect
2. Aorta that overrides the **right** & left ventricles
3. Obstruction to **right** ventricular outflow
4. **Right** ventricular hypertrophy

RIGHT to left shunt
Goals: maintain volume status and SVR (need to ↑ = NEO)

TEF: Tracheoesophageal Fistula
Most common form ends in blind pouch & lower esophagus that connects to trachea – C (then B & E)
Associated with VACERTL syndrome
Principle cause of death = pulmonary complications
↑ secretions No PPV prior to intubation

Pyloric Stenosis:
Infants: projectile vomiting & visible peristalsis
Adults: Peptic ulcer scarring
Hyperchorlemic metabolic alkalosis- ↓Cl, K, Na, Ca
↑ Aldosterone secretion
Cleft palates & esophageal reflux
Tx: surgical pyloromyotomy
Anesthetic Considerations: correct electrolytes, **rehydration**, OG
Post op complication- respiratory depression

Down's syndrome (trisomy 21) Concerns:
1. Difficult intubation: use small than usual ETT
   Large tongue
   Short neck
   Small mouth
2. Neck flexion- Cervical spine dislocation- antlanto-occipt
   Instability & Weak ligaments
3. Congenital heart disease (40% incidence)

Omphalocele vs Gastroschisis:

	Omphalocele	Gastroschisis
Incidence	1 in 5,000	1 in 15,000
Location	Base of umbilicus	Lateral to umbilicus
Anomalies	Yes – Cardiac	No
Presence of Sac	Yes	No
		Infection, hypothermia & Hydration more of a concern

Both: decompress stomach with tube, no nitrous, hydrate 8-16ml/kg/hr, if PIP > 25-30 H_2O – no primary closure

Epiglottitis vs. Croup

	Epiglottis	Croup
Importance	Emergency	Non-emergency
Onset	Rapid, 24 hours	Gradual 24-72 hours
Radiograph	Thumb sign (swollen)	Steeple sign (narrow)
Age	1-7 y.o	Most <2 yo
Cause	Haemophilius B- Flu	Cold (viral)
Fever	Yes- HI	Low grade
Respiratory	Inspiratory stridor	Croupy cough "bark" with inspiratory stridor
Tube Size	½ size small w/ leak	½ size smaller
Tx	Ampicillin &/or vaccine before 2 yrs	Epi neb, O_2, cool humid, steroids
Anesthesia	~Do not attempt to visualize glottis ~sedate while sitting	If ↑ $PaCO_2$ –intubate Epi Neb- 2.25% in 3ml NS @ 0.05ml/kg up to 0.5ml/kg q 1-4 hr

Cystic Fibrosis:
Hereditary disease of exocrine glands of resp and GI
Cl^- transport ↓ w/ ↓ Na & H_2O transport
↑ Thickness of secretions= avoid antisialogogues
↑ RV & airway resistance, ↓ VC & exp flow rate
No ketamine 2nd to increase secretions

Necrotizing enterocolitis:
Most common surgical emergency in the neonate
Premature infants @ > risk for developing
Decreased mesenteric blood flow = ischemia → intestinal mucosal injury
*Foreign body aspiration is most common pediatric surgical emergency

Encephalocele, Meningocele, Myelomeningocele:
Neural Tube Defect- abn fusion of embryologic neural groove
Encephalocele: herniation of brain & meninges through defect in skull producing a fluid filled sac
Meningocele: hernia protrusion of a saclike cyst of meninges filled w/ CSF & no neurological deficits-
Myelomeningeocele: (spina bifida) hernial protrusion of saclike cyst containing meninges, CSF & portion of spinal cord
LATEX precautions & Positioning is key

Inguinal hernia:
Nerves blocked: ilioinguinal and iliohypogastric

Glucose:
Hyperglycemia is neuroprotective for global ischemia in NN
Hypoglycemia can be more detrimental

Tonsillectomy:
↑ Blood loss- 4ml/kg
3 complications: bleeding, laryngospasm & emesis

NPO Guidelines

Substances	Minimum Fasting Hours
Solid Food	8
Commercial Formula/Milk	6
Breast Milk	4
Clear Liquids	2

Mendelsons Syndrome:
Volume > 0.4ml/kg or pH <2.5

Regional Anesthesia:
Spinal & caudal most common
CSF volume x2 of adults = dilution of LA = ↑ dosing & ↓ DOA
Spinal: L3-L4 – distance ~ 1 cm
High spinal – decreasing oxygen saturation #1 sign
Caudal: tip of coccyx to fix the midline & sacral cornua on either side of sacral hiatus
Volume determines height of block
1.2-1.5ml/kg analgesia T4-T6- concentration no > 2.5 mg/kg
Epidural Blood Patch: 0.3ml/kg
Mepivicaine is not metabolize in the NN

Laryngospasm:
1. CPAP- 10-15cm H_2O
2. atropine 0.02mg/kg
3. Succs 1mg/kg IV or 4mg/kg IM

Maternal Physiological & Physical Changes:
Cardiac:
↑ CO, SV, HR, LVEDV, EF, Femoral venous pressure
↓ PVR, MAP, SBP, DBP, **SVR**, PADP, PCWP
CO ↑ the most immediately post-partum
↑ Renin & angiotensin II x12

Respiratory:
↑ IRV, TV, IC, DS, MV (↑ 50%), AV, O_2 consumption (↑ 33%), PaO2 > 100, ↓ MAC, RR (10%)
↓ ERV, RV, **FRC** (20%), chest wall excursion, chest wall, FEV_1 compliance Respiratory Alkalosis CO_2 = 30, restrictive lung
↑ Alveolar vent & ↓ FRC = ↑ inhalation agent uptake & ↓ MAC

GI:
Metabolic Acidosis – HCO_3 = 20
↑ Gastric emptying time & ↓ LES → 2nd to ↑ progesterone
↑ Gastrin production

Hepatic:
Albumin ↓ 4.5 g/dL to 3.3 g/dL = ↓ COP by 5mmHg
↓ Plasma cholinesterase

Renal:
↑ RBF & GFR (50%) → ↓ BUN & serum creatinine
↑ Kidney size & weight & ureters and renal pelvis dilates
Glucosuria & proteinuria

Hemostasis:
Relative hemodilutional maternal anemia,
Hgb > 14 w/PIH or HTN = low volume
↑ Blood volume, plasma volume, RBC, Hbg 11.5 Hct 35.5
↑ Factors 1, 7, 8, 9, 10,12 , Fibrin degradation products, plasminogen
Shortened or ↓: PT, PTT, AT, Factors 11 & 12
No Δ 2 & 5, platelets, bleeding time
PT, PTT, and bleeding times that are greater than twice the normal value represents a high-risk state that should be corrected prior to initiating a neuraxial anesthetic.
Platelet < 100,000 too low to perform a neuraxial anesthetic

Anesthetic considerations w/ Δ:
Smaller ETT ↑ risk of failed intubation
Avoid intranasal manipulations ↑ Minute ventilation required
Do not hyperventilated = ↓ maternal alkalosis = ↓ uterine blood flow
GA = RSI 8 wk. gestation to 6 week of postpartum
2nd to venodilitation = ↑ risk of accidental epidural vein puncture
VA off after delivery to promote uterine tone
Appendectomy is the MOST common surgical procedure
Most common cause of maternal death during GA = Hemorrhage
↓ MAC & ↑ sensitivity to LA

Pharmacology:
H_2 antagonist & or metoclopramide & nonparticulate
Uterine relaxation: nitroglycerin 50-500 mcg IV or MAC 1.5-2.0
DO NOT give esmolol to PIH/HELLP parturient
Maternal alkalosis & fetal acidosis = ↑ fetal ion trapping

β2: relaxes smooth muscles & stops contractions via activation of adenylyl cyclase = ↑ cAMP = myometrial relaxation
S/E: cerebral vasospasm, chest pain, nausea, **pulm edema**, tremor
Ritodrine—issues: hypokalemia, hyperglycemia, tachycardia
Terbutaline – more likely to have arrhythmias

Tocolytics
Methergine**:** exerts uterine effect @ α adrenergic receptors, 0.2mg/IV or intramyometrially
Hemabate: ↑ prostaglandins F2 (safer than meth) 0.25 IM/metral)

Uterine Blood Flow 800ml/min (10% maternal CO)
1. ↓ **Perfusion pressure**: supine, hemorrhage/hypovolemia, HoTN, contractions, sz, Valsalva
2. ↑ Uterine Vascular resistance: catecholamines, vasopress

↑ UBF from regional: pain relief, ↓ SNS activity, ↓ hyperventilation
↓ UBF from regional: HoTN, absorbed LA, unintentional IV inject
*Maternal BP is **only** factor influence **blood flow through placenta**
*uterine blood flow is NOT autoregulated
*α adrenergic receptors predominate uterine vasculature

Greatest risk to fetus from Maternal issues:
Severe hypoxia, HoTN, acidosis

Three layers in placental membrane:
1. Fetal trophoblasts
2. Cytotrophoblasts
3. Syncytiotrophoblasts

Placental Exchange:
1. Diffusion: small ions, respiratory gases, most anesthesia drugs under 1000 Daltons cross,
2. Bulk flow, how water crosses the membrane
3. Active transport and transfer of AA, vitamins, calcium, & iron,
4. Pinocytosis, larger molecules such as immunoglobulins cross
5. Breaks in the placental membrane

Placental Transfer of Drugs:
1. Concentration gradient
2. Molecular weight of drugs – MW < 500 = easier transfer
3. Lipid solubility- lipids soluble drugs = easier crossing
4. Ionization – ions inhibited
VA transfers easily muscle relaxations DO NOT

Maternal supine hypotensive syndrome:
Compression inferior vena cava = ↓ venous return = ↓ SV & Hotn
S/S: diaphoresis, N/V, Δ in cerebration, dizzy, vertigo, ↑ HR, apprehension
Tx: LUD (i.e Right hip up) 15^0

Definitions:
First stage: Regular uterine contractions until cervix fully dilated.
Latent- Cervical effacement 2-3 cm
Active – frequent contractions 3-5 mins & dilatation is 4 cm in a primiparous patient & 3 cm in a multiparous patient.

Second stage: From the end of the first stage until the delivery of the baby is completed.

Third stage: From the delivery of the baby until the placenta and the membranes are expelled.

Polyhydramnios: excessive amniotic fluid around unborn infant

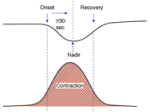
Pain Pathways for Stages of Labor
First stage: uterine contractions & cervical dilatation. Pain travels via **visceral afferent** fibers accompanying sympathetic nerves. Enter cord at T_{11} & $T_{12} \rightarrow T_{10}$, T_{11}, T_{12}, & L_1 spinal segments.
C-Fibers

Second stage: Caused by distention of lower vagina, vulva, and perineum. Pain travels via the Pudendal nerves and enters the cord at S_2, S_3, & S_4 sacral segments. (T_{10}-S_4 dermatomes)

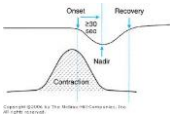
Third Stage: delivery of baby until placenta & membranes out

Paternal Pain Agents in Parturient
Meperidine: Most commonly used opioid – longed DOA- give in early labor (4 hours before delivery) 10-25 mg IV, 25-50 mg IM
Fentanyl: maternal respiratory depression- 50-100mcg/hr.
Butorphanol: partial agonist- no resp depression 1-2 mg IV/ IM
Nalbuphine: partial agonist- no resp depression 10-20 mg IV/ IM
Promethazine: can be used with Demerol – 50-100mg IM
Ketamine: 10-15mg IV- no LOC, up to 1 mg/kg (> = fetal depress)
Naloxone directly to the neonate at a dose of 0.1 mg/kg IM
NO NSAID: they inhibit uterine contractions & promote closure of the fetal DA.

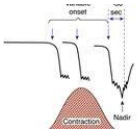
Fetal Heart Tracings:
Early Decelerations (Type 1)
Etiology: head compression or stretching of neck during uterine contractions.
Start & end w/ contraction & UNIFORM
Mild decrease in FHR < 20 BPM



Late Decelerations (Type II)
Etiology: uteroplacental insufficiency & fetal compromise w/ ↓ HR
Onset: begin or near end of contraction
Uniform in appearance + or - variability
SEVERE: if FHR > 45 BPM



Variable Decelerations:
Etiology: cord compression
Nonuniform with variable waveform
Typically associated with fetal asphyxia when: > 60 bpm FRH, duration > 60 seconds or pattern persists > 30 mins
Severe: FRH ↓ 60 by BPM, FHR < 60 BPM or decels 60 secs + Fetal bradycardia & late decelerations = fetal hypoxia (asphyxia)



Obstetrics & Regional Anesthesia
SAB ↓ dose by 25% d/t ↓ epidural space 2nd venous congestion &/or progesterone-induced sensitivity

Most common S/E = Hotn

The diagnosis of abruption placentae with or without fetal distress is a contraindication to regional anesthesia

Paravertebral lumbar sympathetic block: used during 1st stage of labor when traditional neuraxial analgesia is contraindicated

Chloroprocaine: quickly metabolized no opportunity to accumulate in any significant amount.

Ropivacaine: 1/3 less motor block than bupivacaine & causes less CNS & cardiac toxicity, 0.08-0.15% can be administered via an epidural catheter at a rate of 8-12 mL/hour to provide continuous analgesia during labor

T4 level sufficient for **caesarean section**

Injuries:
Dorsiflex to foot= common peroneal nerve
Loss of sensation to lateral thigh = lateral femoral cutaneous
Most common nerve injury during ab hyster= femoral nerve
Nerve injury most common w/ vaginal delivery= lumbosacral

SAB Doses: Lido 5% 50mg/ml (50-75mg) T_7-C_8
Bupivacaine 0.75%- 7.5 mg/ml (12-15mg) T_5-C_8
Tetracaine 1% 7-10mg in D_{10}- hyperbaric T_7-T_5

Epidural Doses: 1.5-2% Lidocaine
0.5% bupivacaine
3% chloroprocaine

Spotty spinal: 2.5mg diazepam up to 10mg IV
Fentanyl 1µg/kg IV
40% N_2O
0.25mg/kg ketamine IV
10-20ml 0.5% lidocaine intraperitoneally

Regional Opioids

	Intrathecal	Epidural
Morphine	0.5-1mg	7.5-10mg
Meperidine	10-20mg	100mg
Fentanyl	10-25mcg	50-100mcg
Sufentanil	3-10mcg	10-30mcg

Pudendal block
Placed through sacrospinous ligament- via Transvaginal approach
2nd stage of labor
Risks: hitting the scalp

Paracervical Block
Anesthetizing nerve fibers innervate uterus, cervix, upper vagina
Submucosally in the vagina @ 3 & 9'oclock
1st stage of labor
Risks: Fetal Brady (33%)

Signs of Fetal distress:
Repetitive late decelerations
Loss of beat to beat variability associate w/ late or deep decels
Sustained FHR < 80 BPM
Fetal Scalp pH < 7.2
Meconium-stained amniotic fluid
IUGR
Oligohydramnois

Maternal Death
Most common (most to least)
Hemorrhage, embolism, preeclampsia, infection & cardiomyopathy

Uterine Atony
Inadequate uterine contractions following delivery
Risk factors: multiple gestations, fetal marosomia, prolonged/rapid labor, tocyotlitcs, VA, retained placenta, chorioamniotis
S/S: vaginal bleeding and boggy uterus
TX: bimanual compression, uterine massage, IV pic, hysterectomy

Chorioaminoitis
One of the most common infections during pregnancy
Based on S/S Temp > 38 = CARDINAL symptom

Uterine Cord Prolapse:
May lead to fetal hypoxia
Dx: sudden fetal bradycardia or profound decels w/ physical exam
Tx: immediate steep trendelenburg or knee to chest position

Antepartum Hemorrhage:
Placenta Previa:- No Pain
The partial or total covering of the cervical os by the placenta
First bleeding episode – usually preterm w/ no contractions
Painless vaginal bleeding during 2nd or 3rd trimester
Marginal, total, partial- cesarean section
Avoid vaginal exams
↑ incidence of asymmetric IUGH

Placental Abruption: Pain
Separation of the placenta from deciduas basalias before delivery
Acute bleeding from exposed vessels= lot of BLOOD
Coagulopathies: ↓ plts, factor 5 & 8, ↑ fibrin split products
Major complications: hemorrhagic shock, ARF, coagulopathy
One of the most common causes of fetal demise.
Definitive Tx: delivery of fetus

Abnormal Placental Implantation
Placenta accrete: placenta adheres to the surface of the myometrium w/o invasion or passage through uterine muscle
Most common indication for hysterectomy
Placenta increta refers to placental implantation within the myometrium (confined to myometrium)\=
Placenta percreta refers to the condition where the placenta completely penetrates the myometrium or other pelvic structures

Amniotic Fluid Embolism: AFE 3rd cause of death
Amniotic fluids into maternal circulation
S/S: tachypnea, cyanosis, shock, generalized bleeding, ↓ CO, acute PE, uterine atony, ARDS, arrhythmias, bleeding, SZ
Tx: aggressive cardioplulamony resuscitation, stabilization, fast delivery,

Uterine Rupture:
Uterine wall defect resulting in fetal distress or maternal hemorrhage
Risk Factors: prior C-section, uterine scar, trauma, forces, hard labor
Most diagnostic sign = fetal distress- loss of fetal heart tones
Continuous abdominal pain & Hotn (↑ in VBAC)
Tx: volume resuscitation & immediate laparotomy & hyster poss

Pregnancy Induced Hypertension: (↑ thromboxane A_2)
HTN = ↓ CI & ↓ blood volume
Gestational hypertension: HTN w/o edema or proteinuria
Preeclampsia: HTN w/proteinuria and edema during pregnancy.
Eclampsia: preeclamptic patient w/ sz
HELLP: form of preeclampsia characterized by hemolysis, elevated liver enzymes, and a low platelet count.
*uterine blood flow decreases
* avoid ketamine
↓ PCHE – 33% eclampsia & 60% in HELLP = careful dose of Succs

S/S Severe PIH:
BP ↑ 160/110
Proteinuria ↑ 5 g/day
Oliguria, < 500 ml/day
Pulmonary edema
Hepatic tenderness or HELLP syndrome
CNS manifestations: HA, visual disturbances, or seizures

Complications of PIH:
Cerebral hemorrhage- #1 death
Pulmonary edema- #2 death
Renal failure
Cerebral edema
DIC
Airway obstruction

Complications that necessitate delivery:
SBP >/= 160 or DBP >/= 110 – 24-48 hrs
Progressive renal dysfunction
progressive thrombocytopenia
Liver dysfunction
premonitor signs of eclampsia
Evidence of fetal jeopardy

Actions of Magnesium:
Beneficial: Anticonvulsant, vasodilatation, ↑ UBF, ↑ RBF, ↑ prostacyclin, ↓ ACE, ↓ renin activity, tocolytic & bronchodilitation
Detrimental: ↓ FH variability, myoneural blocking effects, generalized muscle weakness, ↑ sensitivity to MR
MOA: ↓ presynaptic release of AcH & ↓ post synaptic sensitivity preeclamptic patients & works @ NMDA → ↓ SVR & ↑ CI.

IV load = 4g over 15-20 mins → 1-4g/hr IV gtt

$MgSO_4$ Levels:(mEq/L) to convert to mg/dl Divide by 0.8

1.5-2	Normal
4-8	therapeutic
5-10	long PR, wide QRS
10	lose DT reflex
15	respiratory paralysis, SA & AV block
20	CV arrest

Magnesium: drug of choice sz prophylaxis in PIH
Tx for Mg overdose : Calcium Gluconate

Hydralazine: useful in pregnancy—mainstay
↓ BP
↑ Uteroplacental BF

Heart Disease in the Parturient:
Group One: mitral valve dz, aortic insufficiency, L to R shunt
Regional Anesthesia: especially continuous epidural
NEO
Group Two: AS, R to L shunt, primary pulm HTN
Regional anesthesia CONTRAINDICATED

VAE in the parturient:
Place in slight reverse trendelenburg w/ L lateral tilt of 15^0

Four findings that suggest DIC:
1. Thrombocytopenia 2. Prolonged PT
3. Prolonged PTT 4. ↓ Serum fibrinogen, ↑ Fibrin split products

DIC is associate with 3 OB problems:
1. retention of dead fetus
2. Placental abruption
3. AFE

Lab Tests for DIC:

	Normal OB	DIC
Plasma fibrinogen	400-650 mg/dL	<150 mg/dL
Platelet count	150,000-300,000/mm^3	<50,000/mm^3
Thrombin time	15-20 sec	>100 sec
Prothrombin time	10-12 sec	>100 sec
Partial thromboplastin	35-50 sec	>100 sec
Fibrin split products	<16 mcg/ml	>200 mcg/ml
Red blood cell fragment	No	Yes

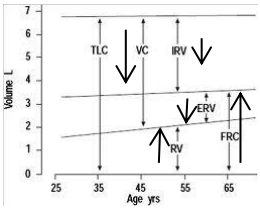
Geriatrics
Geriatrics/Elderly = > 65y.o Ages = >80y.o
Progressive loss of functional reserve in ALL organ systems
1 % decline each year after 30 years old

Homeostasis: Able to maintain; however, less able to restore when trauma, disease or drug

A-A gradient: >/= 20 (nml 8)

Surgical Risk:
1.age 2. Patient status
3.Coexisiting dz 4. Elective vs. emergent & type of surgery

Geriatric System Changes
Cardiac: fx declines 50% (20-80 y.o)
*Increased***:** left ventricular wall thickness, LVH d/t chronic ↑ afterload, left ventricular wall tension (law of Laplace), afterload, cardiac workload 2nd to ↓ aortic compliance - AC, SBP, PVR, Circulation time, conduction fibrosis, dysrhythmias, SA node cell loss, Vagal tone – d/t ↓ sensitivity of adrenergic receptors, systolic HTN d/t ↑ PP 2nd to vessel stiffness,
Decreased: cardiac reserve, CO –d/t ↑ in AL, CI, HR, left ventricular compliance, chronotropic & inotropic responses, baroreceptor fx, adrenergic sensitivity
No Change: DBP, resting systolic fx, excitation-contraction coupling, ionized Ca levels, contractile proteins, SV

Respiratory: Restrictive


Increased: vocal cord stimulation for closure, airway obstruction, risk of aspiration, pulmonary complications, physiologic dead space, WOB, potential for hypoxia, FRB, Closing volume and Closing capacity, alveolar compliance, resp depression w/opioids, collagen
P_aCO_2-P_ACO_2 gradient d/t V/Q mismatch, $V_{D/}V_{T\backslash}$
Decreased: elastin fibers, tissue elasticity, lung recoil, ability to cough, chest wall compliance – kyphosis, VC, ERV, IRV, response to hypoxia & hypercarbia, protective reflexed, cervical spine & TMJ mobility, ease of mask ventilation, PaO_2 (0.4mmHg/yr. after 20)
No Change:$PaCO_2$, P_AO_2

CC=FRC @ 44 y.o CC > FRC @ 66yo
Closing volume is 10% of VC in the young & 40% in the elderly
P_AO_2-PaO_2 gradient = 0.21 (age +2.5)
$PaO_{2=}$ 102-Age/3

Endocrine:
Increased: Insulin resistance, heat loss
Decreased: heat production, hypothalamic temp regulation

GI/Hepatobiliary:
Increased: gastric pH, AAG-1 (↑ binding of basic-LA, opioids)
Decreased: liver mass, HBF, liver fx, biotransformation, albumin production, PCHE – MEN, gastric emptying, plasma clearance
No change: Hepatocellular fx

Nervous System:
Increased: skeletal muscle atrophy, degeneration of peripheral nerve cells, Threshold –proprioceptor, hearing, temp thresh, touch thresh, vision thresh, CSF, cerebral cortex neuron loss, **SNS**
Decreased: CBF, intracranial volume, **gray matter**, brain mass, skeletal muscle steadiness-strength-control, conduction velocity, **↓response to β**
No change: Autoregulation

Renal:
Increased: BUN, ADH response to hypertonic saline load (caution w/fluids), ability to develop hyper/hypo kalemia
Decreased: kidney mass, RBF d/t ↓CO (50%), renal plasma flow, GFR, renal fx, muscle mass, creatinine production, fluid handling (prone to fluid overload), Na handling, concentrating ability, response to ADH, response to aldosterone
No change: serum creatinine
*Creatine Clearance is most sensitive indicator of renal fx in elderly 70ml/min @ y.o
Thermoregulation: > 80 shivering @ 35^0 vs. younger pt. at $36.1C^0$

Pharmacology:
Increased: circulation time, body fat, Vd for lipids, recovery of VA
Decreased: muscle mass, body head production, core body temp, basal metabolic requirements, MAC, Total body water, Vd for water soluble drugs, dosing for barbs-- opioid antagonists-- benzos
Doses:
Increase: Atropine, Isoproterenol & other Beta agonists
Decrease:
Thiopental- ↓ 15% induction - 2.1mg/kg- / same main
Propofol - ↓ 20% - 1.7 mg/kg Etomidate: ↓0. 2mg/kg
Midazolam- ↓ 50% 0.02-0.03mg/kg Opioids: ↓ 50%
Remifentanil ↓50-70% Pan, Vec & Roc - ↓20%
Succs- slightly reduced dose
Beta receptor responsiveness 2nd to ↓ receptor affinity & alterations in signal conduction - ↓ cAMP
No Changes: atracurium, neostigmine, Edrophonium

Anesthetic Considerations:
Positioning and padding of importance d/t fragile skin, ↓ sub q fat, & poor skin turgor. -- Collage loss & ↓ elasticity- sensitive – muscoskeletal limitations- some positioning not possible

Preoperative concern: Heart> renal> hepatic > pulm > multi- DM
Postoperative concern: Heart & lungs

CAD & COPD = CXR

IV induction = SLOWER Inhalation Induction= FASTER

Postoperative delierum
*Higher Risk for delirium and Postoperative cognitive dysfunction
1-3 days post op- resolves in hours to days
@ risk: > 70 yrs, hx of delirium, etoh abuse, narcotic
Greatest w/Orthopedic procedures
Haldol – benzos make worse- unless ETOH related

Geriatric & Regional
↓ Cmin for LA

SAB	**Epidural**
↑DOA	↓ DOA
↑Sensory block	↓ Motor Block
↓ Dose	↓ segment dose
	↑ volume cephalad spread

T8 for TURPS
Cysto SAB @ T10

5 Similarities of Neonates to Geriatrics:
1.↓ ability to ↑ HR 2nd to hypovolemia
2.↓ arterial O_2 tension
3. impaired ability to cough
4.↓ renal tubular function
5.↑ susceptibility to hypothermia

Progeria
Aka: Hutchinson-Gilford Syndrome- premature aging
Average death 13 y.o
Airway effects: mandibular hypoplasia, micrognathia, glottis opening is narrow
Anesthesia concerns: organ systems fx, positioning

Alzheimer's
Deposition of amyloid beta peptides produces neuritic plaques & neurofibrillary tangles= ↓ NT fx & death of neurons.
The limbic system and cortex are most affected.
Dx:only postmortem examination of brain tissue is definitively diagnostic of Alzheimer's
*Risk Factors***:** Family history, age greater than 65, poor education, history of SZ, head injury, MI, & hypothyroidism.
Tx: Cholinesterase inhibitors: rivastigmine, donepazil, & Galantamine
S/E: nausea, vomiting, bradycardia, syncope, & fatigue.

Anesthesia/Pharm:
Acetylcholinesterase inhibitors may have a ↑ DOA w/succs
Anticholinergic – Use glycopyrrolate is blood brain
May be resistance to NDMR d/t use of acetylcholinesterase inhibiting drugs

Obesity
Values:
BMI < 18.5 = Underweight 18.5-24.9 = Normal
25-29.9 = Overweight 30-39.9 = Obesity
40-49.9 = Extreme obesity 50-59.9 = Superobesity 60+ = Super-super obesity

>28 BMI or 20% >35 BMI (45%> IBW) morbid obesity

IBW = ht in cm – 100 (m) or 105 (f) 1cm=2.54 in
BMI = kg/m^2

BMI > 35- 40 is cut off for ambulatory surgery center

Cardiac:
Increased: CO, Blood Volume (50ml/kg), incidence of HTN
EKG Δ's : Low QRS voltage, LVH criteria, left atrial enlargement, t-wave flattening inferior & arterial leads, & prolonged QT
OSA ♥ effects: polycythemia, pulmonary & systemic vasoconstriction, ↑ risk for ischemic heart dz & cerebrovascular dz, & RHF, cardiomegaly, CHF
Extra 0.1 L/min of CO for each additional kilogram of fat.

Respiratory:
Increased: Diaphragm, O2 consumption, CO_2 production, WOB, RR, Hypoxemia
Decreased: ERV, IC, FRC, VC, PaO_2, chest wall compliance,
No Change: PFT's, ling compliance
Miscellaneous: Right to Left shunt , TV within CC, **Restrictive**

Hepatic/GI:
↑ GERD, cholelithiasis, pancreatitis & nonalcoholic fatty liver dz
↑ liver enzymes , ↑ glucose, cholesterol
Renal:
Increased: GFR renal tubular resorption,
impaired Na^+ excretion = worse HTN
No change: RBF

Endocrine:
Hyperinsulinemia- activates SNS = Na retention = ↑ HTN

Anesthesia Management:
Neck circumference single best predictor of difficult airway
BMI of 50 = RSI w/ cricoid pressure
↑ Brachial plexus injury
Phen-phen use – inquire about valve issues
Appetite suppressant – SSRI's = catecholamine depletion

Regional:
↓ LA dose by 20% 2nd vascular engorgement of the epidural space, the level and onset of an epidural block can be unpredictable

Pharmacology:
↑ Vd for meds

Drugs distributed mainly to lean tissue dosed on LBW & include: Thiopental, propofol, rocuronium, vecuronium, atracurium, midazolam cisatracurium, fentanyl, sufentanil, & remifentanil

Drugs distributed to lean & adipose dosed on TBW & include: Succinylcholine and Dexmedetomidine

Hetastarch- 20ml/kg – based on IBW

OSH (pickwickian)	**OSA**
Obesity hypoventilation syndrome = males & females	obstructive sleep apnea males > females
awake PaCO2 > 45 mmHg, doesn't exhibit nocturnal airway obstruction unless concomitant OSA pulmonary hypertension	normal awake $PaCO_2$ sleep induced obstruction
somnolence, sleep apnea, hypercapnia, & hypoxemia	normal $paCO_2$, pH, pulm compliance
↓ alveolar ventilation, cyanosis Polycythemia, enlarged heart, Hypoxemia – PaO_2 < 65mmHg Rales	

Liposuction
1ml of wetting solution per 1ml of fat. >5000 ml – fluid overload concern may promote- hypoxemia, HTN, & pulm edema
Wetting solution: used to emulsify fat, provide anesthesia, create hemostasis – solution is dilute epi 1:100,000 & lidocaine 0.05-0.1%
PE is # 1 concern = 25% of liposuction deaths

Positioning
Lateral Position
↑ risk of rhabdo
Awake & spont breathing: dependent lung better V & Q
Anesthetized & spont breathing: non-dep better V, depen- better Q
Anesthetized & mech vent:↑ non dep V, depen better Q = ↑V/Q
Most compromised position in anesthetized & mech vent patient

Ax roll/chest roll – to protect brachial plexus = relief of pressure from axillary neurovascular bundle & prevent ↓ blood flow to hand

Postoperative tonsillectomy patient with head down

Lithotomy
↓ FRC
Respiratory embarrassment – Pulm edema, Restrictive lg dz
5 nerves that may be injured
Common Peroneal (most common)
Sciatic saphenous
Femoral Obturator (excess flexion)

Prone
Upright to prone = ↓SV, CO, FRC
Supine to prone = ↑ FRC

Trendelenberg position
↑ MAP, PCWP, SVR, venous return, CVP, ICP
↓ CI, oxygen delivery, O_2 consumption, CO
↓ TLC, VC, FRC
***FRC** & pulmonary compliance MOST
West Zone III
↑ Pulm edema

11 Complications of steep trendelenburg
1.Hotn 2. Blood loss/hypovolemia
3. VAE 4. Ocular complications
5. venous thrombus 6. ETT migration
7. atelectasis 8. Neuropathy
9. arthralgia 10. Finger injuries
11. regurgitation

Sitting
90^0 = sitting 45^0 = beach chair
Hotn is most frequent complication
Air embolism most likely with this position
Least likely to have ↑ ICP
Neurosurgery 4 reasons to have head up: provides better access, improves venous drainage, lowers ICP, easier to observe face

Avoid excessive cervical flexion = impedance of arterial & venous blood flow—Hypoperfusion of the brain- macroglossia
*Maintain at least 2 fingerbreadths between chin & sternum

Lawn chair ↓ sciatic nerve injuries

Prone
Large breast = positioning medial and cephalad

vision loss during prone procedures in Ischemic optic neuropathy and central retinal artery occlusion account for 89% of cases of postoperative vision loss in prone

Pneumocephalus
Caused because air enters the cranium while the pt. is in a head up position at a time when the volume of intracranial contents has been reduced as a result of some combo of ↓ CO_2, good venous drainage, osmotic diuresis, CSF loss from field.

Manifests: delayed emergence from GA, severe HA

Bladder perforation
Awake- shoulder pain
Anesthetized – tachy, htn or hotn

Total parotidectomy
spares the facial nerve, while a radical parotidectomy removes the facial nerve

Hyperextension of neck = compression of ipsilateral and/or contralateral vertebral arteries

Bone Cement:
Methylmethacrylate toxicity = sudden hotn
MOA – transient fat/air embolism from bone marrow or causes vasodilatation & ↓ SVR
Minimized MMA toxicity: hydrate & ↑ FiO_2

Radial Prostectomy- complications
1. Hemorrage (#1) 2. Compromised vent fx
3. Air embolism 4. Obturator nerve injury
5. DVT

8 causes of rhabdomyolysis
1. Major crush injury 2.thermal/electrical injury
3. Acute muscle ischemia d/t arterial occlusion
4. Acute muscle injury 2nd prolonged immobilization
5. Compartment syndromes 6.MH
7. Extreme lithotomy 8. Hyperlordotic position

6 Congenital syndromes associated with difficult intubation
1.Downs 2. Goldenhar 3.klippel-fiel
4. Pierre Robin 5. Treacher Collins 6. Turner

Ludwig's Angina
Overwhelming generalization septic cellulites of submandibular region. Usually after dental procedures. S/S: chills, fever, drooling, ability to open mouth, difficulty speaking. Caused by hemolytic streptococci.
Airway management: preliminary tracheotomy using LA in awake patient = Safest

Contraindications for Cricothyrotomy
Children < 6 years old
Laryngeal fractures

Scleroderma
Severe jaw limitation
Decreased compliance = difficult ventilation
Arterial hypoxia 2nd decreased diffusion of O2 across alveoli
No RSI – awake intubation with head up position= safer

Preoperative Assessment
Mouth opening – 40mm -- 2 fingerbreadths
Thyroidmental distance - > 6.5 cm nml < 6= difficult airway
Silent MI – HTN & DM
ETOH – Hypomagnesium, hypokalemia, metabolic alkalosis

Inappropriate for Ambulatory Surgery:
1. Major Surgery/ Major blood loss
2. ASA 3& 4
3. Morbidly obese
4. Complex pain management
5. Fever, wheezing, nasal congestion, cough, URI
6. Patients susceptible to MH
7. Uncontrolled sz activity
8. Active substance abuse
9. Active infection
10. Uncooperative or unreliable patients
11. No responsible adult @ home during convalescence

Medications to Hold before surgery
Oral Glycemic – am of surgery
Diuretics- am of surgery

Postoperative complications
Postoperative pulmonary complications: thoracic and upper Abdominal surgery = highest risk.
.
Hypoxia in the PACU: hypoventilation & ↑ R to L intrapulmonary shunting 2nd to ↓ FRC (#1)

HoTn 2nd to #1- Hypovolemia HTN 2nd #1 Pain

#1 Postoperative Complication – PONV
High risk PONV = children, women, previous hx, hx of motion sickness, anxiety, abd, gyn , laparoscopic, opioids, surgical duration

Carotid Endarterectomy
$PaCO_2$ 35-45 mmHg – Avoid ↓ CO_2 → vasoconstriction
Stump pressure transmitted pressure through the circle of willis
< 50 = shunting
Avoid hyperglycemia
Maintain BP > preop BP (MAP 80-100)
Bradycardia d/t surgical baroreceptor stim = lidocaine by surgeon
Post-operative HTN = carotid sinus baroreceptors

Causes of morbidity & mortality #1- MI & #2-Stroke

Heparin 5,000-10,000 & Protamine 0.5mg/100 u of Hep

Monitoring: EEG, somatosensory evoked potentials, internal carotid artery stump pressure, and transcranial Doppler. During awake carotid endarterectomy, the patient's neurologic status and responsiveness is the indicator of cerebral perfusion

Nerves: Smile: facial nerve, say "EEE": superior and recurrent laryngeal nerves, shrug shoulders: spinal accessory nerve, swallow: glossopharyngeal, stick his tongue out: hypoglossal nerve

Nerve Injury: phrenic, vagus, recurrent and superior laryngeal nerves, ansa hypoglossi, and hypoglossal nerves

AAA
Prior to clamping- 5000U Heparin IV, PAP 5-12mmHg
Above clamp: ↑ EDV, ESV, SVR, MAP, ↓LVEF, CO
Below clamp: ↓SVR, MAP, tissue perfusion
Anaerobic metabolism & lactate accum
Release of Clamp: ↓SVR, BP, CO 2nd volume shifts- central hypovolemia = ↓ venous return
Maintain intravascular volume!
Thoracic artery = 11% spinal cord ischemia & 6% paraplegia
Ventilation: hyperventilate, and CPAP to nondep
Classified: Debakey & Stanford

Transplant
Organs most frequently transplanted: kidney & liver
6 brainstem reflexes absent in brain death:
1. Pupillary response to light
2. Corneal reflex
3. Oculocephalic reflex (dolls eyes)
4. Oculovestibular reflex (caloric response)
5. Gag & cough reflex
6. Facial motor response

Goals of anesthetic management of organ donors:
*Maintain euvolemia
1. SBP > 100mmHg
2. PO_2 > 100mmHg
3. Urine Output > 100ml/hr
4. Hbg concentration 100g/L
5. CVP 5-10mmHg
6. FiO2 < 40% (if tolerated)
7. Glucose < 200mg/dl
8. Peak airway pressures < 30mmHg

Absolute contraindications: active infection

Intraoperative renal considerations:
1. SBP > 90mmHg 2. MAP > 60mmHg
3. CVP > 10mmHg

Liver Transplant
Severe acidosis – THAM – trishydroxymethl

Pneumoperitoneum
Hemodynamic Δ's
↑SVR & MAP , ↓ CI Healthy patient – no Δ in CO, HR
5 pulmonary fx Δ's
↑PIP ↓VC ↓FRC
↑Ppl ↓respiratory system compliance

Eye Surgeries
Normal IOP: 10-22mmHg based on rate of aqueous humor formation & rate of aqueous humor outflow
3 parameters: ***CVP***, arterial blood pressure, arterial partial pressure
IOP ↑ with hypercarbia IOP ↓ with hypocarbia
Strabismus: cardio effects of ocular meds, oculocardiac reflex, MH, PONV
N_2O-avoid *10days* postop after use of sulfur hexafluoride

OLV
Vent management: F_IO_2 @ 1.0, TV 10-12ml/kg, $PaCO_2$ 35-40
Hypoxia: #1 – check position of DLT w/FOB, CPAP, PEEP
CPAP- the single most effective means to increase PaO2 w/OLV
CPAP- non-dependent lung 5-10cm H_2O
PEEP- dependent lung 5-10cm H_2O
Other actions: periodically inflate collapsed lung, ligate PA
Greatest risk = hypoxemia
↓ PaO_2 2nd to right to left intrapulmonary shunt
Tubes: White- Right, Carlens, Left, Robertshaw- both (no hook)

Inhibition of HPV
1. Hypocapnia 2. Vasodilators 3.VA
4. High or low pulmonary artery pressure
5. High or low mixed venous partial pressures
6. Pulmonary infections
Shunting of the blood away from non-dependent lung in OLV

Nasal Intubation
1. Temporomandibular surgery
2. Lefort I or II osteotomy for surgical repositing of maxilla
3. I & D of severe oral facial infections
4. Intermaxillary fixation

TRAM
Avoid vasopressors and N_2O during abd closure

Mediastinoscopy
Surgical procedure where a mediastinoscope is inserted into the mediastinal space in order to view and biopsy lymph nodes.
Monitors: Art-line & pulse ox Right BP cuff – Left
The most common reason for doing this *bronchogenic carcinoma*
Compression: innominate or Right brachiocephalic
Complications: #1- hemorrhage & #2pneumothorax tearing of great vessels, chylothorax,bronchospasm from airway manipulation, air embolism, arrhythmias, & esophageal laceration.

Mesenteric Traction Syndrome
Hypotension, tachycardia, cutaneous hyperemia, and hypoxia are signs caused by the release of vasoactive amines (principally prostacyclin) from the vascular bed of the mesentery. As a result, serum prostaglandin levels increase substantially

Tx: H1 & H2, Ketorolac, Neo

Tourniquet
Inflated: Arm 50mmHg > SBP (250mmHg)
Leg 100mmHg > SBP (300mmHg)
Tourniquet times should not exceed 1.5 to 2 hours
↓CVP & SBP
↑ HR, ETCO2, PaCO2, serum K, & serum lactate levels

Restrictive Lung Disease
Extrinsic: pectus carinatum, pectus excavatum, kyphosis, scoliosis, and flail chest as well as obesity, neuromuscular disorders
Intrinsic: Sarcoidosis

Bowel Obstruction
metoclopramide should never be administered to a patient suffering from a bowel obstruction due to the possibility of perforation from its prokinetic effects

Nerve Conduction Pathways
Paleospinothalamic tract (medial spinothalamic tract) projects to the medial thalamic nuclei - associated with autonomic & emotional responses to pain.
spinocervical tract possesses fibers that ascend to the lateral cervical nucleus and then cross to the contralateral thalamus.
spinomesencephalic tract projects to the midbrain reticular formation and may generate nondiscriminatory pain sensations.
Neospinothalamic tract (lateral spinothalamic tract) and sends fibers to the posterior nuclei of the thalamus- location and intensity of pain.

Burns
Types: Chemical, electrical, thermal, and inhalation
Classification: first, second, third, or fourth degree.
First-degree consists of erythema w/ only microscopic damage to the superficial epidermis.
*Second degree (*partial thickness), extend through the epidermis into the dermis. Spontaneous regeneration of the skin is possible
Third-degree: total destruction of the skin, dermal appendages, & epithelial elements occurs with no spontaneous regeneration of the skin possible.
Fourth-degree: involve muscle, fascia, and bone.

Fluids: first 24 hours is % body surface area X Kg X 2 to 4. (hct)

Rule of Nines: adults: each arm 9%, each leg 18%, the entire trunk is 36%, head 9%, perineum 1%.

Hct & viscosity: ↑ significantly
Electrical: ↑ renal issues maintain UO 1-1.5ml/hr
UO: Adults 0.5ml/kg/hr < 30kg peds 1ml/kg/hr
Upper airway inhalation injury: superheated air & steam.
The larynx, epiglottis, tongue, and pharyngeal tissue swell rapidly = complete airway obstruction.

Lower airway inhalation injury: soot and chemical toxins.
The toxins produce acidic & alkaline compounds disrupt the capillary permeability of pulmonary epithelium = alveolar damage.

CO: > 15 = toxic tx = 100% O_2

Anesthesia: OR 28-30 ^{0} C, most heat loss from burns = evaporation
Resistant to NDMB d/t ↑ # of cholinergic nicotinic receptors

Trauma
Most common cause of coagulopathy = Dilutional thrombocytopenia
Basal skull fx : blood behind ear drum, CSF from nose & ears
No drugs that ↑ ICP: ketamine, succs, N_2O

Cardiogenic Shock	**vs.**	**Hypovolemic shock**
↑ PAOP > 15mmHg, ↓CI, ↑ SVR		↓ PAOP, nml CI, nml or ↑ SVR

Autonomic Hyperreflexia
65-80% ↑ T7 (unlikely below T10)
Triggering stimuli: bladder, or bowel distention, heat/cold, uterine contractions, pyelonephritis
S/S: Hallmark – HTN & reflex bradycarda
Below level of injury – SNS activation = vasoconstriction & HTN
Above level of injury- SNS blockade = vasodilatation

Pain- Neurolytic blocks
Indicated for cancer patients with incurable, intractable cancer pain
Not permanent- may last 2-6 months
Celiac plexus block for pancreatic cancer = most effective
Blocked with alcohol or phenol
Most common: celiac plexus, lumbar sympathetic chain, hypogastic plexus and ganglion impar
Greater occipital nerve block = relief from cluster headaches

Pain fibers from the head are carried within the trigeminal, facial, glossopharyngeal, and vagus nerves.

Corticosteroids are the most common multipurpose coanalgesic used for cancer pain

Post-operative shivering medications:
1. Clonidine
2. Physostigmine
3. Serotonin antagonists
4. Propofol.

Electricity:
Macroshock = large voltage applied to skin/tissue
Microshock = small voltage/current directly to the ♥
V-fib caused by 50microamp-micro/100-2500 milliamp- macro

Grounded in OR: Power supply, Patient, The floor

Units & Measurements
PO_2 = 760mmHg
1mmHg=1.36cm H_2O
1atm = 760mmHg=14.7 psi = 101 kPa = 1 bar
1 psi =54 mmHg

% = g/100 ml
mcg/ml = 1,000,000/denominator
Concentration= Amount/Volume
Amount = concentration x volume

CO_2 Absorber
Baralyme = 80% $Ca(OH)_2$ + 20% $Ba(OH)_2$ (no silica)
10.2L of CO_2/100g
Soda Lime = 94% $Ca(OH)_2$ + 5% NaOH + 1% KOH
26L of CO_2/100g
Forms: $CaCO_3$ + NaOH (2KOH) → neutralization reaction
CO_2 = exothermic
4-8 mesh = optimal surface area
Silica = ↑ hardness & ↓ dust
Exhausted when exposed to carbonic acid – Ethyl Violet

CO2 absorber: soda lime & carbon dioxide occurs 2 steps.
#1 CO2 + H2O --> H2CO3 (intermediate product of reaction)
#2 H2CO3 + 2NaOH --> Na2CO3 + 2H2O + heat. (final)

E cylinder: O_2 = 660 Liters, 2200 psi

Air = 625 Liters, 1800 psi Law of Laplace
T=P x r

N_2O = 1590 Liters, 750 psi

Woods Metal plug melts @ 200^0 F (71.4C). and 3000 psig
Woods Metal = Tin, Cadmium, Lead, Bismuth

When N_2O < 745 = no more liquid & ~ 400L of N_2O remains

Equipment
DISS- Quick connectors indexed for specific gas- SAFETY & is a check Valve- Wall Hose to machine – pressor of 40-50psig
free floating value on back of gas machine
at pressure of 1380 kPa (200 psi) or less.
Closes if cylinder pressure is on & pipeline pressure is off
Hanger Yoke-
Pinks for E cylinders – **O_2 pin index 2,5 & N_2O is 3,5**
Diaphragm Valve : 1st & 2nd regulators -fx is to reduce pressure
1st stage- 40-50psig (intermediate)
2nd stage- 40-50 to 16 psig
O_2 Flush Valve- 35-75L/min, 40-50 psig (intermediate)
Pressure Sensor Shut-Off Valve: (only senses PRESSURE)
Senses O_2@ 50 psig, shuts of N_2O if O_2 pressure falls
O_2 pressure @ 25psig to keep open
Needle Valve for waste scavenging: flow (not suction or vacuum)

Gas flow during breathing:
Inspiration: inhalation check valve & soda lime, through soda lime- to CGO and from rebreathing bag
Expiration: though exhalation check valve to APL/rebreathing bag

Components of the High Pressure system of AM : > 55psig
1.hanger yoke
2. yoke block w/ check valves (free floating)
3. cylinder pressure gauge
4. cylinder pressure regulators

Components of intermediate pressure system of AM: 40-50 psig
1. Ventilator power inlet
2. pipeline inlets, check valves, pressure gauges
3. flow meter valves
4. oxygen pressure- failure devices
5. oxygen second stage regulators
6. flush valve

Components of the low-pressure system of AM: 16 psig
1. flow meter tube
2. vaporizers
3. check valves
4. CGO

Tec 6 vaporizer – dual circuit, gas vapor blender
39^0 C & pressurized to 2 atmospheres (1300mmHg)
@ high elevation – need to ↑ concentration to raise PP

Oxygen supply failure alarm
Sets off an alarm if the oxygen pressure falls below a standard set by the manufacturer typically **30 psig**
Must engage within **5 seconds** of the disconnect and cannot be disabled by the anesthetist

Prevent hypoxia from lack of oxygen flow, but does not prevent the flow of anesthesia gases (still possibly allow a hypoxic mixture of gases to be delivered).

Pumping Effect:
An intermittent back pressure caused by positive pressure ventilation or use of the oxygen flush valve results in ↑ vaporizer output. (not happening on newer machines)
Low: flows, vapor dial settings, levels of anesthetic in vaporizer
High: RR & PIP

Tipping of the vaporizer = most agent to the patient
Breathing Systems:
Open System- no mask on the face ie. NC
Semi-Open: mask on face- no rebreathing = ↑ FGF
Semi-Closed: mask on face- some rebreathing w/ regular flows
Closed: Mask on Face: - complete rebreathing APL closed & low flows (**150-500ml/min physiological requirements)**

Mapleson Systems:
Prevention of rebreathing – **spont ventilation**: A> DFE > CB
Prevention of rebreathing **Controlled vent:** DFE > BC > A
All Mapleson can ventilate the apneic patient
Mapleson D modified = Bain → best for controlled vent
Mapleson E only one w/o reservoir bag
Mapleson F = Jackson Reese - peds d/t ↓ WOB but heat loss

Forced Air Warmer
Maximum temp 48 C
Average contact 46 C

Blood Salvage Contraindications:

Infection	malignant cells
Urine	bowel contents
Amniotic fluid	

Heat & Moisture Exchanger:
Conserves some exhaled water & heat–return to pt. in inspired gas.
Bacterial/viral filtration: Hydrophobic > Hydroscopic
Contraindicated: copious secretions, uncuffed endotracheal tube

LMA
Contraindicated: risk for aspiration, fixed ↓ pulmonary compliance, unable to understand instruction, hiatal hernia

Pacemaker
Chamber paced A,V,D
Chamber sensed A,V,D
Response – **I**nhibit, **T**rigger, **D**ouble, **O**-none

Pulse Oximetry:
Beer Lambert
Fx w/ 2 wavelengths: Red light 660 nm – Deoxygenated Hbg
Infrared light 940nm- Oxygenate Hbg
Fingers are one of the last places to showing desat (central first)

Changes in Pulse Oximetry
False High: Caboxyhemoglobin & methemoglobin, **severe anemia**
SpO_2 < 85%
False Low: Methylene blue, prominent venous pulsations, injection if dyes- indigo carmine, lymphazurin, nitrobenzene, indocyamine green, patent blue

Capnography:



5 monitors that can detect disconnection
1.pulse oximetry
2. mass spectrometer
3. capnography
4. stethoscope
5. spirometer

Mass Spectrophy:
Gases analyzed: CO_2, O_2, N_2 & inhaled agents (Not PaO_2)
Gas sampled is ionized by an electron beam and passed through a magnetic field.

Raman: measures gas concentrations by analyzing the intensity of light emitted when a gas sample returns to an unexcited state after being energized by a laser beam

Infrared: they detect gases based on the amount of infrared light that is absorbed by the sample- **can not detect oxygen content**
Most popular in the OR & based on beer lambert

Isoproterenol = high false measurements

Peripheral Nerve Stimulator -- CURRENT
TOF – 2Hz – two twitches per second or one every 0.5 seconds
Tetany- 5 seconds @ 50Hz
DBS-2 trains of 3 impulses @ 50Hz – separated by 750 ms (best to determine fade)
Post Tetanic- 50 Hz x 5 seconds – 3 second pause – stim @ 1 Hz
of visible post tetanic twitches correlates inversely to time required for return of single twin or train-of-four responses
Ulnar nerve = adductor pollicis- abdomen
Facial nerve = orbicularis oculi – laryngx
Twitches:
1. one visible twitch = 95% blockade
2. two visible twitch= 80-85% blockade
3. three visible twitches= 75-80%
4. four visible twitches = < 75%

BIS – based on HYPNOSIS
100- Awake
90-70- Light/Moderate Sedation
70-60- Deep sedation (low probablility of recall)
60-40- general anesthesia
40-10- deep hypnotic state
10-0- flat line EEG
< 40 reflect burst suppression.

Artifact can be produced by: Output from cardiac pacemakers, muscle activity such as twitching, shivering, or blinking, incorrect electrode placement, & high frequency electrical devices such as IV pumps and warming devices can all produce artifact.

Paradoxical delta waves may occasionally be seen during anesthetic maintenance and emergence.

Blood Pressure Cuffs: 40% of arms circumference
Overestimation if cuff: too loose, too small, or positioned below the level of the heart

Pacemakers
Most common indications = SSS & complete ♥ block
Identification codes
1st = chamber paced
2nd= chamber sensed -- eletrocautery affects
3rd= response to sensing
4th= programmability rate modulation } AICD
5th antitachyarrythmic fx

Medications
Atropine – 0.5-1mg x3 Isoproterenol
Succs can inhibit the pulse generator d/t fasiculations
Avoid nitrous if implanted within 1-2 days ago

Magnet
Pacer- converts to a fixed rate (asynchronous) mode
AICD- disables it – loud continuous high pitch sound

Sterilization
Disinfection: destroys most microorganisms- except SPORES
Sterilization: all viable forms of microbial life – yes to spores
7 chemical disinfectants: quaternary ammonium, alcohols, glutaraldehydes (Cidex), hydrogen-peroxide, formaldehyde, phenolic compounds, chlorine (bleach)
Ethylene oxide- for objects that can't be heated in steam autoclave
Glutaraldehyde (Cidex) & hydrogen peroxide = SPORES destruct
Quaternary ammoniums – WILL NOT kill m tuberculosis

Lasers
Monochromatic (a single wavelength),coherent (it oscillates in the same phase),collimated (it is a narrow, parallel beam of light).
Lasers can be both long (CO2laser) and short (YAG laser) wavelengths.
Regulates the manufacture & marketing of medical lasers FDA

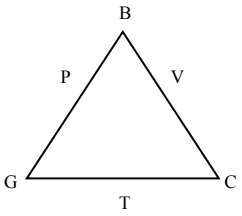
Major risks: Thermal injury, eye injuries, electrical hazards, fire, transmission of viruses, and contaminants in the smoke plume

Nd-YAG laser – OD(optical density) 5 or > for 1,064 nm/Green

N-95 respirator mask: Laser vaporization of condylomatous lesions d/t release toxic chemicals: benzene & formaldehyde as well as viable viruses capable of transmitting the disease

ETT: PVC most susceptible to ignition by CO2 lasers
Red Rubber – wrapped with metallic tape – deflect beam
Silicone- if hit with laser- vaporizes & can aspirate parts
Inflate with sterile saline & methylene blue

Gas Laws
"Can These Girls Possibly Be Virgins"



Boyle = $P_1V_1 = P_2V_2$
Charles = $V_1/T_1 = V_2/T_2$
Gay-Lussac = $P_1/T_1 = P_2/T_2$
(Temp on bottom)

Avogadro's Hypothesis = 1 mole of gas @ STP = 22.4 Liters
STP = C and 1 atm
Avogadro's Number
6.023×10^{23} = 1 mole
1 mole = 0^0 = 1 atm = 760 mmHg = 22.4L
Gases liquefy if: (1) sufficient pressure is applied and
(2) temperature is below critical temperature
N_2O = 39.5 ^{0}C
O_2 = -119 ^{0}C

Fick Diffusion: $\frac{(P_1 - P_2)\ (\text{Area})\ (\text{Solubility})}{(\text{Membrane thickness})\ (\sqrt{}\ \text{Molecular Weight})}$

***the concentration gradient is the most important factor in determining the rate of diffusion of a drug across a membrane

% Concentration = (Partial Pressure/Atm) X 100

Partial Pressure = $\frac{\%\ \text{Concentration} \times \text{Atm}}{100}$

Partial Pressure H_2O @ 37^0 C = 47 mm Hg
~Concentration effect
~2nd gas effect
~Diffusion hypoxia
~N_2O = ↑ V or ↑ P in gas spaces

Laminar Flow: Poiseuilles
$F=\pi r^4 \Delta P/8nl$ Flow
F= flow, r = radius, n= viscosity, l= length
Doubling the radius = 16x the flow
Tripling the radius = 81x the flow
***Δ in radius = most dramatic effect on flow
Angle < 25 degrees = laminar flow

Reynolds #- turbulent flow
Re= v x d x e/ n
v = fluid velocity, d= diameter, p=density, n = viscosity
>1,500-2000 = turbulent flow
***Density (p) determines flow when turbulent flow present

Venturi/Bernoulli
Flow through constricted region of the tube = ↑ Flow & corresponding ↓ in pressure in area of narrowing
Venturi O_2 mask, nebulizer, jet ventilation
Anterior Leaflet in IHSS

LeChatelier's: Law of mass action
↑ concentration of reactant→ reaction to ↑ products
↓ concentration of reactant→ reaction to ↑ reactant (↓ products)

Henry's Law
Calculation of dissolved O_2 & CO_2 in blood
O_2 – 0.003ml/100ml blood/ mmHg
O_2= % FiO_2 x5 = PaO_2
CO_2 – 0.067 ml/100ml blood/ mmHg

Dalton's Law of Partial Pressure
Total pressure in a mixture is = to the sum of pressures of each gas
$P_{total} = P_1 + P_2 + P_3 \ldots\ldots P_N$
O_2 – 160 mmHg (21%)
N_2 – 600 mmHg (79%)
Total – 760 mmHg

Joule-Thompson
A cylinder cools and condenses after opening a valve – Joule is cool

Beer Law
The intensity of light is altered as transmitted through liquid.
The intensity of the light falls exponentially as light passes through the liquid.
~Pulse oximetry & Infrared absorption spectroscopy
Infrared → oxygenated Hbg = wavelength 940 nanometers
Visible red→ deoxygenated → wavelength 660 nanometers

Law of Laplace
T= P x r – cylindrically shaped structures thus ↑T = ↑ r
Applies to: blood vessels, left ventricle (Frank-Starling)
T= (P x r) /2 -- spherically- shaped structures thus ↑r = ↑ T, ↓r = ↑ P
Applies to alveoli – ARDS

Humidification
Relative Humidity (%) = $\frac{\text{actual vapor pressure}}{\text{Saturated vapor pressure}} \times 100$

↓ T = ↓ capacity to hold H_2O → H_2O condenses

Partial pressure of saturated water vapor @ 37^0 = 47mmHg

Ohms Law
SVR

Ostwald Solubility Coefficient
Solubility of a gas in a fluid
Amount of gas in solution is **inversely** proportion to temp
Hypothermia = ↑ solubility of VA in blood & tissue

Van Der Wals:
Non-ideal gas behavior

Inverse Square Law
Distance from source = amount of exposure

Three Steps of QA programs:
1. Define the norm
2. Are adverse events caused by deviations from norm?
3. Prevent adverse events

Fires in OR:
Components needed for Fire: fuel, oxygen & ignition source
Steps if fire occurs:
1. Stop ventilation
2. Stop O_2 Flow
3. Extubate patient
4. Extinguish the fire
5. Mask ventilate
6. Reintubate

History of Anestehsia:
Agatha Hodgins→ formed the AANA in 1931
Alice Magaw → 'Mother of Anesthesia
Cocaine → first LA
Sister M. Bernard→ first nurse anesthetist in Erie, PA

Legal:
A writ of mandamus= an order by a court to force a party to commence some required action.

Stare decisis= refers to the doctrine of common law in which courts adhere to the prior decisions of other courts.

Res ipsa loquitor= (the thing speaks for itself) refers to an event that would not have occurred 'but for actions of the defendant'.

Tort = Civil Wrongdoing

Negligence = Failure to use reasonable care, which is that level of care recognized as acceptable and appropriate given the circumstances.

Replevin= is a lawsuit to recover goods improperly taken by another.

Battery= intentionally causing harmful or offensive contact with a person or to something close to them
Assault= intentionally causing the apprehension of an immediate and harmful contact

Four Elements of Malpractice:
1.Duty- prove defendant had a duty to the plaintiff
2.Breach of Duty- prove defendant failed to fulfill duty to defedant
3. Causation - reasonably close relationship proven to exist between breach of duty by defendant and the injury that resulted
4.Damages - prove some injury occurred due to the breach in duty

Informed Consent Includes:
1. Risks
2. Benefits
3. Complications
4. Alternatives

Standard Agencies

Department of Transportation: defines compressed gas standard
Interstate Commerce Commission: sets specification for compressed gas cylinder construction
Federal Food, Drug & Cosmetic Act: defines compressed gas standard
United States Pharmacopeia: develops purity standards for gases
National Fire Prevention Association: recommendations for the construction and location of BULK oxygen containers
Compressed Gas Association: sets standards of practice
American National Standards Institute (ANSI): performance & safety requirements for components of the AM, ETT, connectors,vacuum & gas pressure regulators
Food and Drug Administration: promulgates standards for medical devices and gases
Joint Commission: voluntary accrediting agency
American Society for Testing Materials (ASTM): assess technology & revises standards
National Institute of Occupational Safety & Health: standards to protect the health & safety of workers

Type I error When researchers, on the basis of a statistical test, erroneously concludethat there is evidence of an association between two variables when, in fact, there is not
Type II error When researchers, on the basis of a statistical test, erroneously conclude that there is evidence of no association between two variables when, in fact there is